Commun Nonlinear Sci Numer Simulat 80 (2020) 104927

Contents lists available at ScienceDirect

Commun Nonlinear Sci Numer Simulat

journal homepage: www.elsevier.com/locate/cnsns

Research paper

Focusing and defocusing Hirota equations with non-zero boundary conditions: Inverse scattering transforms and soliton solutions

Guoqiang Zhang[a], Shuyan Chen[b,c], Zhenya Yan[b,c,*]

[a] Department of Mathematical Sciences, Tsinghua University, Beijing 100084, China
[b] Key Laboratory of Mathematics Mechanization, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China
[c] School of Mathematical Sciences, University of Chinese Academy of Sciences, Beijing 100049, China

ARTICLE INFO

Article history:
Received 6 February 2019
Revised 13 July 2019
Accepted 25 July 2019
Available online 27 July 2019

Keywords:
Hirota equation
Non-zero boundary conditions
Inverse scattering
Riemann-Hilbert problem
Solitons
Breathers

ABSTRACT

In this paper, we investigate the inverse scattering transforms and soliton solutions of both focusing and defocusing Hirota equations with non-zero boundary conditions (NZBCs). The inverse problems are solved via the study of the matrix Riemann-Hilbert problems. As a consequence, we present the general solutions for the potentials, and explicit expressions for the reflectionless potentials. Moreover, the trace formulae and theta conditions are also given. Particularly, we discuss the simple-pole and double-pole solutions for the focusing case, and the simple-pole solutions for the defocusing case. Moreover, the results of the focusing case with NZBCs can reduce to ones of the focusing case with zero-boundary condition (ZBC). These obtained solutions are useful to explain the related nonlinear wave phenomena.

© 2019 Elsevier B.V. All rights reserved.

1. Introduction

Since the inverse scattering transform (IST) was first presented to solve exactly the well-known Korteweg-de Vries (KdV) equation with a Lax pair [1], the IST has been used to study more and more physically important nonlinear wave equations with Lax pairs such as the nonlinear Schrödinger (NLS) equation, modified KdV equation, Benjamin-Ono equation, sine-Gordon equation, Kadomtsev-Petviashvili equation, Davey-Stewartson equation, and etc. (see, e.g., Refs. [2–4] and references therein). As physically interesting models, the NLS equation and its extensions appear in many fields of nonlinear science such as nonlinear optics, Bose-Einstein condensates, deep ocean, and even finance (see, e.g., Refs. [5–14] and references therein). The focusing ($\sigma=-1$) and defocusing ($\sigma=1$) NLS equations

$$iq_t+q_{xx}-2\sigma(|q|^2-q_0^2)q=0,\quad \sigma=\pm1,\quad q_0\in\mathbb{R} \tag{1}$$

with ZBC, $\lim_{x\to\pm\infty}q(x,t)=0$ and $q_0=0$, and NZBCs, $\lim_{x\to\pm\infty}q(x,t)=q_\pm$ with $|q_\pm|=q_0\neq0$, have been solved via the standard IST and its extension, bilinear method, and dress method (see, e.g., Refs. [8,15–30] and references therein). Recently, the ISTs of integrable nonlinear wave equations with NZBCs have been paid much attention to based on the solutions of the related Riemann-Hilbert problems (see, e.g., Refs. [22–30] and references therein).

* Corresponding author. Tel.: +861062651370.
E-mail address: zyyan@mmrc.iss.ac.cn (Z. Yan).

https://doi.org/10.1016/j.cnsns.2019.104927
1007-5704/© 2019 Elsevier B.V. All rights reserved.

To study more complicated nonlinear wave phenomena such as the phenomena of oceanic and optical waves, the extensions of the NLS equation were needed. For example, when the optical pulses become shorter (e.g., 100 fs [6]), higher-order dispersive and nonlinear effects such as third-order dispersion, self-frequency shift, and self-steepening arising from the stimulated Raman scattering are significant in the study of ultra-short optical pulse propagation [31–33]. The Hirota equation [34] is an important one of the above-mentioned higher-order NLS equations, and is completely integrable.

This paper focuses on the ISTs and solutions for both focusing and defocusing Hirota equations [34] with NZBCs at infinity

$$\begin{aligned} &iq_t+\alpha[q_{xx}-2\sigma(|q|^2-q_0^2)q]+i\beta(q_{xxx}-6\sigma|q|^2q_x)=0,\quad \sigma=\pm1,\ (x,t)\in\mathbb{R}^2,\\ &\lim_{x\to\pm\infty}q(x,t)=q_\pm,\quad |q_\pm|=q_0\neq0, \end{aligned} \tag{2}$$

where the real constants α,β denote the second-order and third-order dispersions, respectively, and $\sigma=-1,+1$ stand for the focusing and defocusing cases, respectively. Recently, the rogue waves of the focusing Hirota equation and its modulational form have been studied via the Darboux transform [33,35–39]. For $\beta=0$ and $\alpha=1$, the Hirota equation with NZBCs (2) reduces to the NLS equation with NZBCs (1), while when $\alpha=0$ and $\beta=1$, the Hirota equation with NZBCs (2) reduces to the complex modified Korteweg-de Vries (mKdV) equation with NZBCs

$$\begin{aligned} &q_t+q_{xxx}-6\sigma|q|^2q_x=0,\quad \sigma=\pm1,\ (x,t)\in\mathbb{R}^2,\\ &\lim_{x\to\pm\infty}q(x,t)=q_\pm,\quad |q_\pm|=q_0\neq0. \end{aligned} \tag{3}$$

Eq. (2) is just a compatibility condition $X_t-T_x=[X,T]=0$ of the system of linear partial differential equations (Lax pair) [39,40]

$$\begin{aligned} &\Phi_x=X\Phi,\quad X(x,t;k)=ik\sigma_3+Q,\\ &\Phi_t=T\Phi,\quad T(x,t;k)=\alpha T_{nls}+\beta T_{cmkdv}, \end{aligned} \tag{4}$$

where $\Phi=\Phi(x,t;k)$ is a 2×2 matrix-valued eigenfunction, $k\in\mathbb{C}$ is a spectral parameter, and

$$\begin{aligned} &T_{nls}=-2kX+i\sigma_3(Q_x-Q^2+\sigma q_0^2),\\ &T_{cmkdv}=-2k(T_{nls}-i\sigma q_0^2\sigma_3)+[Q_x,Q]+2Q^3-Q_{xx}, \end{aligned} \tag{5}$$

$$\sigma_3=\begin{bmatrix}1&0\\0&-1\end{bmatrix},\quad Q=Q(x,t)=\begin{bmatrix}0&q(x,t)\\ \sigma q^*(x,t)&0\end{bmatrix},\quad [Q_x,Q]\equiv Q_xQ-QQ_x, \tag{6}$$

with $q^*(x,t)$ standing for the complex conjugate of $q(x,t)$. Particularly, when $\beta=0$, the Lax pair (4) reduces to one of the NLS Eq. (1), while when $\alpha=0$, the Lax pair (4) becomes one of the complex mKdV Eq. (3).

In this paper, we will study the IST and soliton solutions for the Hirota equation with NZBCs (2) starting from its Lax pair (4) for any parameters α and β with $\alpha^2+\beta^2\neq0$. The rest of this paper is arranged as follows. In Section 2, we investigate the focusing Hirota equation with NZBCs such that its simple-pole and double-pole solutions are found via solving the matrix Riemann-Hilbert problems with the reflectionless potentials. Some representative solitons are given. Moreover, the trace formulae and theta conditions are also discussed. In Section 3, the defocusing Hirota equation with NZBCs is explored such that its simple-pole solutions are obtained by solving the matrix Riemann-Hilbert problem with the reflectionless potential, and their trace formulae and theta conditions are also obtained. Moreover, some representative simple-pole multi-dark solitons are studied. Finally, some conclusions and discussions are given in Section 4.

2. The focusing hirota equation with NZBCs

2.1. The direct scattering with NZBCs

Since the asymptotic problem ($x\to\pm\infty$) of the Lax pair (4) for the focusing ($\sigma=-1$) Hirota equation with NZBCs (2) is given by

$$\begin{aligned} &\Phi_x=X_\pm\Phi,\qquad X_\pm=\lim_{x\to\pm\infty}X=ik\sigma_3+Q_\pm,\\ &\Phi_t=T_\pm\Phi,\qquad T_\pm=\lim_{x\to\pm\infty}T=[\beta(4k^2-2q_0^2)-2\alpha k]X_\pm, \end{aligned} \tag{7}$$

with $Q_\pm=\lim_{x\to\pm\infty}Q(x,t)=\begin{bmatrix}0&q_\pm\\-q_\pm^*&0\end{bmatrix}$, thus Eq. (7) admits the fundamental matrix solutions

$$\Phi_{bg}^f(x,t;k)=\begin{cases}E_\pm^f(k)\,\mathrm{e}^{i\theta(x,t;k)\sigma_3}, & k\neq\pm iq_0,\\ I+\left[x-2(\alpha k+3\beta q_0^2)\,t\right]X_\pm(k), & k=\pm iq_0,\end{cases} \tag{8}$$

where I is a 2×2 unit matrix,

$$E_\pm^f(k)=\begin{bmatrix}1&\frac{iq_\pm}{k+\lambda(k)}\\ \frac{iq_\pm^*}{k+\lambda(k)}&1\end{bmatrix},\quad \theta(x,t;k)=\lambda(k)\left\{x+[\beta(4k^2-2q_0^2)-2\alpha k]t\right\},\quad \lambda^2(k)=k^2+q_0^2. \tag{9}$$

To further investigate the analyticity of the Jost solutions of the Lax pair (4) with $\sigma=-1$, we need to give the regions of Im $\lambda(k)>0\,(<0)$ in the above-defined function $\theta(x,t;k)$ (see, e.g., Ref. [26]) for the focusing NLS case):

Table 1
Conformal mapping from two sheets of k-plane to z-plane, where $\mathbb{Q}_j^k$ ($\mathbb{Q}_j^z$) $j = 1, 2, 3, 4$ denote the first, second, third and fourth quadrants excluding the boundaries of k-plane (z-plane), respectively.

Sheet-I of k-plane	→	z-plane	Sheet-II of k-plane	→	z-plane
$\mathbb{Q}_j^k,\ j=1,2,3,4$	→	$\mathbb{Q}_j^z \bigcap \{\|z\|>q_0\},\ j=1,2,3,4$	$\mathbb{Q}_j^k,\ j=1,2,3,4$	→	$\mathbb{Q}_j^z \bigcap \{\|z\|<q_0\},\ j=1,2,3,4$
$iq_0[0,\ 1]$	→	$\{z = q_0 e^{i\theta_z},\ \theta_z \in \pi[0,\ 1/2]\}$	$iq_0[0,\ 1]$	→	$\{z = q_0 e^{i\theta_z},\ \theta_z \in \pi[-1,\ -1/2]\}$
$iq_0[-1,\ 0]$	→	$\{z = q_0 e^{i\theta_z},\ \theta_z \in \pi[1,\ 3/2]\}$	$iq_0[-1,\ 0]$	→	$\{z = q_0 e^{i\theta_z},\ \theta_z \in \pi[-1/2,\ 0]\}$
$\pm iq_0[1,\ \infty)$	→	$\pm iq_0[1,\ \infty)$	$\pm iq_0[1,\ \infty)$	→	$\pm iq_0[1,\ 0]$
$[0,\ \pm\infty)$	→	$\pm q_0[1,\ \infty)$	$[0,\ \pm\infty)$	→	$\pm q_0[-1,\ 0]$

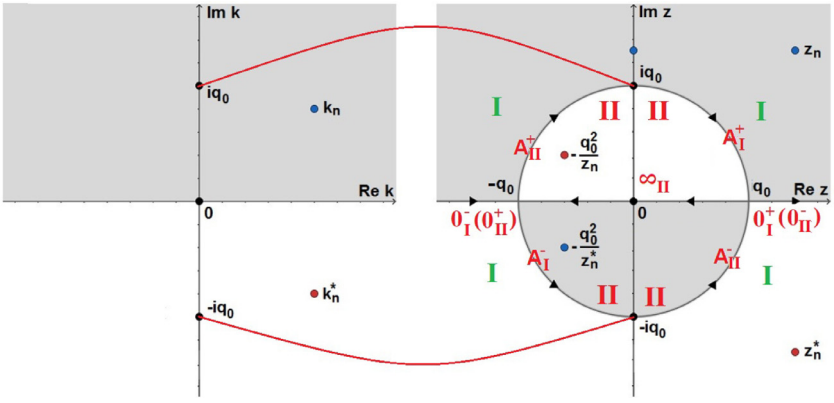


Fig. 1. The grey (white) region for $\mathrm{Im}\,\lambda > 0$ ($\mathrm{Im}\,\lambda < 0$) in distinct spectral planes of the focusing Lax pair with NZBCs. Left: the first sheet of the Riemann surface, showing the discrete spectrum; Right: the complex z-plane, showing the discrete spectrum [zeros of $s_{11}(z)$ (blue) in the grey region and those of $s_{22}(z)$ (red) in the white region], and the orientation of the jump contours for the related Riemann-Hilbert problem. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

- For $q_0 \neq 0$, that is, the boundary conditions (BCs) are NZBCs, $\lambda(k)$ satisfying $\lambda^2(k) = k^2 + q_0^2$ in the complex plane is a doubly branched function of k with two branch points being $k = \pm iq_0$ and the branch cut being the segment $iq_0[-1, 1]$. Let $k \pm iq_0 = r_\pm\, e^{i\theta_\pm + 2im_\pm\pi}$ $(r_\pm > 0,\ \theta_\pm \in [-\pi/2,\ 3\pi/2),\ m_\pm \in \mathbb{Z})$, then two single-valued analytical branches of the complex k-plane are defined by Sheet-I: $\lambda_{\mathrm{I}}(k) = \sqrt{r_+ r_-}\, e^{i\frac{\theta_+ + \theta_-}{2}}$ and Sheet-II: $\lambda_{\mathrm{II}}(k) = -\lambda_{\mathrm{I}}(k)$. In this case, one can introduce a uniformization variable z defined by the conformal mapping [3]: $z = k + \lambda(k)$, whose inverse mapping is $k(z) = \frac{1}{2}(z - \frac{q_0^2}{z}),\ \lambda(z) = \frac{1}{2}(z + \frac{q_0^2}{z})$. The mappings between the two-sheeted Riemann k-surface and complex z-plane are presented in Table 1 (see also Fig. 1).
- For $q_0 = 0$, the NZBCs reduces to the ZBC, in which $\lambda(k)$ satisfying $\lambda^2(k) = k^2$ in the complex plane is also a doubly branched function of k with two branch points being $k = 0$ and $k = \infty$ and the branch cut chosen as the segment $[0, +\infty)$. Thus, we have $\lambda_{\mathrm{I}} = k$ and $\lambda_{\mathrm{II}} = -k$ for $\mathrm{Arg}(\lambda_{\mathrm{I}} \in [0, 2\pi))$. we have $\mathrm{Im}\lambda_{\mathrm{I}}(k) = \mathrm{Im}k > 0\,(< 0)$, i.e., the upper-half (lower-half) plane of k-plane, and $\mathrm{Im}\lambda_{\mathrm{II}}(k) = -\mathrm{Im}k > 0\,(< 0)$, i.e., the lower-half (upper-half) plane of k-plane. Moreover, $\lambda(k) \in \mathbb{R}$ only for $k \in \mathbb{R}$.

Let $\mathbb{A} = iq_0[-1, 1]$ with $\mathbb{A}^+ = iq_0[0, 1]$ and $\mathbb{A}^- = iq_0[-1, 0]$, $C_0 = \{z \in \mathbb{C} : |z| = q_0\}$, and

$$\begin{aligned} D_+^f &= \{z \in \mathbb{C} : \mathrm{Im}(\lambda(z)) = \tfrac{1}{2}\mathrm{Im}(z)(1 - q_0^2/|z|^2) > 0\}, \\ D_-^f &= \{z \in \mathbb{C} : \mathrm{Im}(\lambda(z)) = \tfrac{1}{2}\mathrm{Im}(z)(1 - q_0^2/|z|^2) < 0\}. \end{aligned} \tag{10}$$

The continuous spectrum of $X_\pm = \lim_{x\to\pm\infty} X$ is the set of all values of z satisfying $\lambda(z) \in \mathbb{R}$ [26], that is, $z \in \Sigma^f = \mathbb{R} \cup C_0$, which are the jump contours for the related Riemann-Hilbert problem. Since $T_\pm = [\beta(4k^2 - 2q_0^2) - 2\alpha k]X_\pm$, that is, $[X_\pm, T_\pm] = 0$, thus, for all $z \in \Sigma^f$, similarly to [26,30] for the focusing NLS equation with NZBCs, one can define simultaneously the Jost solutions $\Phi_\pm(x, t; z)$ of both parts of the Lax pair (4) satisfying the boundary conditions

$$\Phi_\pm(x, t; z) = E_\pm^f(z)\, e^{i\theta(x,t;z)\sigma_3} + o(1), \quad x \to \pm\infty. \tag{11}$$

Based on $\Phi_x = X_\pm\Phi + \Delta Q_\pm\Phi$ with $\Delta Q_\pm(x, t) = Q(x, t) - Q_\pm$ with $Q_\pm = \lim_{x\to\pm\infty} Q|_{\sigma=-1}$, we know that the modified Jost solutions $\mu_\pm(x, t; z)$

$$\mu_\pm(x, t; z) = \Phi_\pm(x, t; z)\, e^{-i\theta(x,t;z)\sigma_3} \to E_\pm^f(z), \quad x \to \pm\infty, \tag{12}$$

admit the forms

$$\mu_\pm = \begin{cases} E_\pm^f(z)\Big\{I + \int_{\pm\infty}^x e^{i\lambda(x-y)\hat{\sigma}_3}\Big[(E_\pm^f(z))^{-1}\Delta Q_\pm(y, t)\,\mu_\pm(y, t; z)\Big]dy\Big\}, & z \neq \pm iq_0,\ q - q_\pm \in L^1(\mathbb{R}^\pm), \\ E_\pm^f(z) + \int_{\pm\infty}^x [I + (x-y)(Q_\pm \mp q_0\sigma_3)]\Delta Q_\pm(y, t)\,\mu_\pm(y, t; z)dy, & z = \pm iq_0,\ (1+|x|)(q-q_\pm) \in L^1(\mathbb{R}^\pm), \end{cases} \tag{13}$$

where $e^{\alpha\hat{\sigma}_3}A := e^{\alpha\sigma_3}Ae^{-\alpha\sigma_3}$.

Let $\Sigma_0^f := \Sigma^f\backslash\{\pm iq_0\}$, $\mu_\pm(x,t;z) = (\mu_{\pm1}(x,t;z),\ \mu_{\pm2}(x,t;z))$, and $\Phi_\pm(x,t;z) = (\Phi_{\pm1}(x,t;z),\ \Phi_{\pm2}(x,t;z))$. Since they contain these terms $e^{\pm i\lambda(x-y)}$, thus according to the properties of these functions in distinct domains and the definition (13) of $\mu_\pm(x,\ t;\ z)$, as well as the relation (12) between $\mu_\pm(x,\ t;\ z)$ and $\Phi(x,t;z)$, we have easily the following property (see [26] for the details):

Proposition 1. *Suppose $q - q_\pm \in L^1(\mathbb{R}^\pm)$, then the Jost functions $\Phi_{\pm2}(x,t;z)$ and the modified forms $\mu_{\pm2}(x,\ t;\ z))$ given by Eqs. (12) and (13) have unique solutions in Σ_0^f. Moreover, $\mu_{+1}(x,t;z)$, $\mu_{-2}(x,t;z)$, $\Phi_{+1}(x,t;z)$ and $\Phi_{-2}(x,t;z)$ can be analytically extended to D_+^f, and continuously to $D_+^f \cup \Sigma_0^f$, whereas $\mu_{-1}(x,t;z)$, $\mu_{+2}(x,t;z)$, $\Phi_{-1}(x,t;z)$, and $\Phi_{+2}(x,t;z)$ can be analytically extended to D_-^f, and continuously extended to $D_-^f \cup \Sigma_0^f$.*

Since Tr $X(x,t;z)$ =Tr $T(x,t;z) = 0$ in the Lax pair (4) yields $(\det\Phi_\pm)_x = (\det\Phi_\pm)_t = 0$, thus one can find

$$\det\Phi_\pm = \lim_{x\to\pm\infty}\mu_\pm = \det E_\pm^f(z) = \gamma_f(z) \tag{14}$$

in terms of Liouville's formula, where $\gamma_f(z) = 1 + q_0^2/z^2 \neq 0$ for $z \neq \pm iq_0$. Since $\Phi_\pm(x,t;z)$ are both fundamental matrix solutions of the Lax pair, thus there exists a constant matrix $S(z)$ between them so that

$$\Phi_+(x,t;z) = \Phi_-(x,t;z)\,S(z), \quad z \in \Sigma_0^f, \tag{15}$$

where $S(z) = \left(s_{ij}(z)\right)_{2\times2}$ with $s_{ij}(z)$'s being called scattering coefficients. It follows from the relation (15) that

$$\begin{aligned} &s_{11}(z) = \gamma_f^{-1}(z)|\Phi_{+1}(x,t;z),\Phi_{-2}(x,t;z)|, \quad s_{22}(z) = \gamma_f^{-1}(z)|\Phi_{-1}(x,t;z),\Phi_{+2}(x,t;z)|,\\ &s_{12}(z) = \gamma_f^{-1}(z)|\Phi_{+2}(x,t;z),\Phi_{-2}(x,t;z)|, \quad s_{21}(z) = \gamma_f^{-1}(z)|\Phi_{-1}(x,t;z),\Phi_{+1}(x,t;z)|, \end{aligned} \tag{16}$$

and $\det S(z) = 1$, where $|\cdot,\cdot| = \det(\cdot,\cdot)$.

According to Proposition 1, it is easily shown that the defined scattering coefficients $s_{11}(z)$ and $s_{22}(z)$ in $z \in \Sigma_0^f$ can be analytically extended to D_+^f and D_-^f, and continuously extended to $D_+^f \cup \Sigma_0^f$ and $D_-^f \cup \Sigma_0^f$, respectively, whereas another two scattering coefficients $s_{12}(z)$ and $s_{21}(z)$ can not be analytically continued away from Σ_0^f in general.

To solve the matrix Riemann-Hilbert problem during the inverse scattering problem, we require that the potential does not possess spectral singularities [41], i.e., $s_{11}(z)s_{22}(z) \neq 0$ for $z \in \Sigma^f$ and $S(z)$ is continuous for $z = \pm iq_0$. Thus the so-called reflection coefficients are defined by $\rho(z) = s_{21}(z)/s_{11}(z)$, $\hat{\rho}(z) = s_{12}(z)/s_{22}(z)$ for $z \in \Sigma^f$.

2.2. The inverse scattering with NZBCs and simple poles

To study the discrete spectrum and residue conditions in the inverse scattering problem, we need to analyze the symmetries of the scattering matrix $S(z)$. Similarly to [26] for the focusing NLS equation, it follows from the previous definitions of $k(z)$ and $\lambda(z)$ that they correspond to the symmetries: $k(z) = k^*(z^*)$, $k(z) = k(-q_0^2/z)$, $\lambda(z) = \lambda^*(z^*)$, and $\lambda(z) = -\lambda(-q_0^2/z)$, thus we have the symmetries of X, T, and θ as

$$\begin{aligned} &X(x,t;z) = \sigma_2X^*(x,t;z^*)\sigma_2, \quad X(x,t;z) = X(x,t;-q_0^2/z),\\ &T(x,t;z) = \sigma_2T^*(x,t;z^*)\sigma_2, \quad T(x,t;z) = T(x,t;-q_0^2/z),\\ &\theta(x,t;z) = \theta^*(x,t;z^*), \qquad\quad \theta(x,t;z) = -\theta(x,t;-q_0^2/z), \end{aligned} \tag{17}$$

where $\sigma_2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ is the second Pauli matrix.

Combining the above-mentioned symmetries, the Lax pair (4), and Eq. (12), we have the following symmetries for $\Phi_\pm$ and $\mu_\pm$ as

$$\begin{aligned} &\Phi_\pm(x,t;z) = \sigma_2\Phi_\pm^*(x,t;z^*)\sigma_2, \quad \Phi_\pm(x,t;z) = \frac{i}{z}\Phi_\pm(x,t;-q_0^2/z)\sigma_3Q_\pm,\\ &\mu_\pm(x,t;z) = \sigma_2\mu_\pm^*(x,t;z^*)\sigma_2, \quad \mu_\pm(x,t;z) = \frac{i}{z}\mu_\pm(x,t;-q_0^2/z)\sigma_3Q_\pm. \end{aligned} \tag{18}$$

By means of Eq. (18) with Eq. (15), we have the symmetries

$$S(z) = \sigma_2S^*(z^*)\sigma_2, \quad S(z) = (\sigma_3Q_-)^{-1}S(-q_0^2/z)\sigma_3Q_+, \tag{19}$$

which lead to the symmetries between $\rho(z)$ and $\hat{\rho}(z)$ as

$$\rho(z) = -\hat{\rho}^*(z^*), \quad \rho(z) = \frac{q_-^*}{q_-}\hat{\rho}(-q_0^2/z). \tag{20}$$

The discrete spectrum of the scattering problem is the set of all values $z \in \mathbb{C}\backslash\Sigma^f$ such that they possess eigenfunctions in $L^2(\mathbb{R})$. Similarly to Refs. [26,28,30] for the focusing NLS equation, for the focusing Hirota Eq. (2), they should satisfy

$s_{11}(z)=0$ for $z\in D_+^f$ and $s_{22}(z)=0$ for $z\in D_-^f$ such that the corresponding eigenfunctions are in $L^2(\mathbb{R})$ from the first two ones of Eq. (16) and the definition (12) of $\Phi_\pm$.

Firstly, we suppose that $s_{11}(z)$ has N simple zeros in $D_+^f\cap\{z\in\mathbb{C}:|z|>q_0,\ \mathrm{Im}\,z>0\}$ denoted by z_n, $n=1,2,\cdots,N$, that is, $s_{11}(z_n)=0$ and $s'_{11}(z_n)\neq 0\,(n=1,2,\ldots,N)$. It follows from Eq. (19) of the scattering matrix that if $s_{11}(z_n)=0$, then $s_{22}(z_n^*)=s_{22}(-q_0^2/z_n)=s_{11}(-q_0^2/z_n^*)=0$, which leads to the set of discrete spectrum as

$$Z^f=\left\{z_n,\ -\frac{q_0^2}{z_n^*},\ z_n^*,\ -\frac{q_0^2}{z_n}\right\}_{n=1}^{N},\quad s_{11}(z_n)=0,\ z_n\in D_+^f\cap\{z\in\mathbb{C}:|z|>q_0,\ \mathrm{Im}\,z>0\}. \tag{21}$$

Since $s_{11}(z_0)=0$ and $s'_{11}(z_0)\neq 0$ are required for $z_0\in Z^f\cap D_+^f$, then it follows from the first expression in Eq. (16) that there exists a norming constant $b_+(z_0)$ such that

$$\Phi_{+1}(x,t;z_0)=b_+(z_0)\,\Phi_{-2}(x,t;z_0). \tag{22}$$

The residue condition of $\Phi_{+1}(x,t;z)/s_{11}(z)$ in the simple zero $z_0\in Z^f\cap D_+^f$ yields

$$\mathop{\mathrm{Res}}_{z=z_0}\left[\frac{\Phi_{+1}(x,t;z)}{s_{11}(z)}\right]=\frac{\Phi_{+1}(x,t;z_0)}{s'_{11}(z_0)}=\frac{b_+(z_0)}{s'_{11}(z_0)}\,\Phi_{-2}(x,t;z_0). \tag{23}$$

Similarly, it follows from $s_{22}(z_0^*)=0$ and $s'_{22}(z_0^*)\neq 0$ for $z_0^*\in Z^f\cap D_-^f$ and the second expression of Eq. (16) that there exists a norming constant $b_-(z_0^*)$ such that

$$\Phi_{+2}(x,t;z_0^*)=b_-(z_0^*)\,\Phi_{-1}(x,t;z_0^*). \tag{24}$$

The residue condition $\Phi_{+2}(x,t;z)/s_{22}(z)$ in the simple zero $z_0^*\in Z^f\cap D_-^f$ yields

$$\mathop{\mathrm{Res}}_{z=z_0^*}\left[\frac{\Phi_{+2}(x,t;z)}{s_{22}(z)}\right]=\frac{\Phi_{+2}(x,t;z_0^*)}{s'_{22}(z_0^*)}=\frac{b_-(z_0^*)}{s'_{22}(z_0^*)}\,\Phi_{-1}(x,t;z_0^*). \tag{25}$$

For convenience, we rewrite Eqs. (23) and (25) as

$$\begin{aligned}
&\mathop{\mathrm{Res}}_{z=z_0}\left[\frac{\Phi_{+1}(x,t;z)}{s_{11}(z)}\right]=A_+[z_0]\,\Phi_{-2}(x,t;z_0),\quad A_+[z_0]=\frac{b_+(z_0)}{s'_{11}(z_0)},\quad z_0\in Z^f\cap D_+^f,\\
&\mathop{\mathrm{Res}}_{z=z_0^*}\left[\frac{\Phi_{+2}(x,t;z)}{s_{22}(z)}\right]=A_-[z_0^*]\,\Phi_{-1}(x,t;z_0^*),\quad A_-[z_0^*]=\frac{b_-(z_0^*)}{s'_{22}(z_0^*)},\quad z_0^*\in Z^f\cap D_-^f,
\end{aligned} \tag{26}$$

which correspond to the symmetries

$$A_+[z_0]=-A_-^*[z_0^*],\quad A_+[z_0]=\frac{z_0^2}{q_-^2}\,A_-\left[-\frac{q_0^2}{z_0}\right],\quad z_0\in Z^f\cap D_+^f, \tag{27}$$

in terms of symmetries (18) and (19), which further lead to

$$A_+[z_n]=-A_-^*[z_n^*]=\frac{z_n^2}{q_-^2}\,A_-\left[-\frac{q_0^2}{z_n}\right]=-\frac{z_n^2}{q_-^2}\,A_+^*\left[-\frac{q_0^2}{z_n^*}\right],\quad z_n\in Z^f\cap D_+^f. \tag{28}$$

The relation $\Phi_+(x,t;z)=\Phi_-(x,t;z)S(z)$ can be rewritten as

$$\begin{aligned}
&\frac{\Phi_{+1}(x,t;z)}{s_{11}(z)}=\Phi_{-1}(x,t;z)+\rho(z)\,\Phi_{-2}(x,t;z),\\
&\frac{\Phi_{+2}(x,t;z)}{s_{22}(z)}=\hat\rho(z)\,\Phi_{-1}(x,t;z)+\Phi_{-2}(x,t;z),
\end{aligned}$$

which lead to

$$\left(\Phi_{-1}(x,t;z),\frac{\Phi_{+2}(x,t;z)}{s_{22}(z)}\right)=\left(\frac{\Phi_{+1}(x,t;z)}{s_{11}(z)},\Phi_{-2}(x,t;z)\right)(I-J_0(x,t;\lambda)),\quad J_0=\begin{pmatrix}0 & -\hat\rho(z)\\ \rho(z) & \rho(z)\hat\rho(z)\end{pmatrix}. \tag{29}$$

Similarly to [26], we also have the asymptotic behaviors for modified Jost solutions and scattering data as

$$\begin{aligned}
&\mu_\pm(x,t;z)=I+O\left(\tfrac{1}{z}\right), && S(z)=I+O\left(\tfrac{1}{z}\right), && z\to\infty,\\
&\mu_\pm(x,t;z)=\tfrac{i}{z}\,\sigma_3\,Q_\pm+O(1), && S(z)=\tfrac{q_+}{q_-}\,I+O(z), && z\to 0.
\end{aligned} \tag{30}$$

According to the modified Jost functions, let the sectionally meromorphic matrix $M(x,t;z)$ be

$$M(x,t;z)=\begin{cases}M^+(x,t;z)=\left(\frac{\mu_{+1}(x,t;z)}{s_{11}(z)},\ \mu_{-2}(x,t;z)\right)=\left(\frac{\Phi_{+1}(x,t;z)}{s_{11}(z)},\Phi_{-2}(x,t;z)\right)\mathrm{e}^{-i\theta(x,t;z)\sigma_3}, & z\in D_+^f,\\ M^-(x,t;z)=\left(\mu_{-1}(x,t;z),\ \frac{\mu_{+2}(x,t;z)}{s_{22}(z)}\right)=\left(\Phi_{-1}(x,t;z),\frac{\Phi_{+2}(x,t;z)}{s_{22}(z)}\right)\mathrm{e}^{-i\theta(x,t;z)\sigma_3}, & z\in D_-^f.\end{cases} \tag{31}$$

Then, according to Proposition 1 for the analyticity of the eigenfunctions, the relation (29) and asymptotic behaviors (30), it follows from Eq. (31) that we have the following proposition for the Riemann-Hilbert problem:

Proposition 2 *(Riemann-Hilbert problem). The matrix function $M(x, t; z)$ satisfies the matrix Riemann-Hilbert problem:*

- *Analyticity: $M(x, t; z)$ is analytic in $(D_+^f \cup D_-^f)\backslash Z^f$;*
- *Jump condition: $M^-(x, t; z) = M^+(x, t; z)(I - J(x, t; z)), \quad z \in \Sigma^f$ with $J(x, t; z) = \mathrm{e}^{i\theta(x,t;z)\widehat{\sigma}_3} J_0$;*
- *Asymptotic behavior: $M^\pm(x, t; z) = I + O(1/z)$ for $z \to \infty$ and $M^\pm(x, t; z) = \frac{i}{z}\sigma_3 Q_- + O(1)$ for $z \to 0$.*

To conveniently solve the above-mentioned Riemann-Hilbert problem, let

$$\eta_n = \begin{cases} z_n, & n = 1, 2, \cdots, N, \\ -\frac{q_0^2}{z_{n-N}^*}, & n = N+1, N+2, \cdots, 2N, \end{cases} \tag{32}$$

and $\widehat{\eta}_n = -\frac{q_0^2}{\eta_n}$. Then $Z^f = \{\eta_n, \widehat{\eta}_n\}_{n=1}^{2N}$ with $\eta_n \in D_+^f$ and $\widehat{\eta}_n \in D_-^f$. Subtracting out the asymptotic behaviors and the simple pole contributions, that is,

$$M_{sp}(x, t; z) = I + \frac{i}{z}\sigma_3 Q_- + \sum_{n=1}^{2N}\left[\frac{\operatorname{Res}_{z=\eta_n} M^+(x, t; z)}{z - \eta_n} + \frac{\operatorname{Res}_{z=\widehat{\eta}_n} M^-(x, t; z)}{z - \widehat{\eta}_n}\right], \tag{33}$$

from both sides of the above-mentioned jump condition $M^- = M^+(I - J)$ yields

$$M^-(x, t; z) - M_{sp}(x, t; z) = M^+(x, t; z) - M_{sp}(x, t; z) - M^+(x, t; z)J(x, t; z), \tag{34}$$

in which, $M^\pm(x, t; z) - M_{sp}(x, t; z)$ are analytic in $D_\pm^f$. Moreover, their asymptotics are both $O\left(\frac{1}{z}\right)$ as $z \to \infty$ and $O(1)$ as $z \to 0$, and $J(x, t; z)$ is $O\left(\frac{1}{z}\right)$ as $z \to \infty$, and $O(z)$ as $z \to 0$. Therefore, the Cauchy projectors $P^\pm[f](z) = \frac{1}{2\pi i}\int_{\Sigma^f} \frac{f(\zeta)}{\zeta - (z \pm i0)}\, d\zeta$ (where the notation $z \pm i0$ represents the limit taken from the left/right of z) and Plemelj's formulae are used to solve Eq. (34) to yield

$$M(x, t; z) = M_{sp}(x, t; z) + \frac{1}{2\pi i}\int_{\Sigma^f} \frac{M^+(x, t; \zeta)J(x, t; \zeta)}{\zeta - z}\, \mathrm{d}\zeta, \quad z \in \mathbb{C}\backslash\Sigma^f, \tag{35}$$

where $\int_{\Sigma^f}$ stands for the integral along the oriented contours shown in Fig. 1(right).

It follows from Eq. (31) that only its first (second) column has a simple pole at $z = \eta_n$ ($z = \widehat{\eta}_n$). Therefore, by means of Eqs. (12) and (26), the residue parts in Eq. (35) can be found as

$$\frac{\operatorname{Res}_{z=\eta_n} M^+(x, t; z)}{z - \eta_n} + \frac{\operatorname{Res}_{z=\widehat{\eta}_n} M^-(x, t; z)}{z - \widehat{\eta}_n} = \left(\frac{A_+[\eta_n]\,\mathrm{e}^{-2i\theta(x,t;\eta_n)}}{z - \eta_n}\mu_{-2}(x, t; \eta_n), \frac{A_-[\widehat{\eta}_n]\,\mathrm{e}^{2i\theta(x,t;\widehat{\eta}_n)}}{z - \widehat{\eta}_n}\mu_{-1}(x, t; \widehat{\eta}_n)\right), \tag{36}$$

For $z = \eta_s$ $(s = 1, 2, \cdots, 2N)$, it follows from the second column of $M(x, t; z)$ given by Eq. (35) with Eq. (36) that

$$\mu_{-2}(x, t; \eta_s) = \begin{bmatrix} \frac{iq_-}{\eta_s} \\ 1 \end{bmatrix} + \sum_{n=1}^{2N}\frac{A_-[\widehat{\eta}_n]\,\mathrm{e}^{2i\theta(x,t;\widehat{\eta}_n)}}{\eta_s - \widehat{\eta}_n}\mu_{-1}(x, t; \widehat{\eta}_n) + \frac{1}{2\pi i}\int_{\Sigma^f}\frac{(M^+J)_2(x, t; \zeta)}{\zeta - \eta_s}\, d\zeta, \quad s = 1, 2, \cdots, 2N. \tag{37}$$

Moreover, it follows from Eq. (18) that

$$\mu_{-2}(x, t; \eta_s) = \frac{iq_-}{\eta_s}\mu_{-1}(x, t; \widehat{\eta}_s), \quad s = 1, 2, \cdots, 2N. \tag{38}$$

The substitution of Eq. (38) into Eq. (37) yields

$$\sum_{n=1}^{2N}\left(\frac{A_-[\widehat{\eta}_n]\,\mathrm{e}^{2i\theta(x,t;\widehat{\eta}_n)}}{\eta_s - \widehat{\eta}_n} - \frac{iq_-}{\eta_s}\delta_{sn}\right)\mu_{-1}(x, t; \widehat{\eta}_n) + \begin{bmatrix} \frac{iq_-}{\eta_s} \\ 1 \end{bmatrix} + \frac{1}{2\pi i}\int_{\Sigma^f}\frac{(M^+J)_2(x, t; \zeta)}{\zeta - \eta_s}\, \mathrm{d}\zeta = 0, \quad s = 1, 2, \cdots, 2N, \tag{39}$$

where δ_{sn} is the Kronecker delta function. System (39) containing $2N$ equations with $2N$ unknowns $\mu_{-1}(x, t; \widehat{\eta}_n), n = 1, 2, \cdots, 2N$ can yield the solutions $\mu_{-1}(x, t; \widehat{\eta}_s), s = 1, 2, \cdots, 2N$ such that one can also get $\mu_{-2}(x, t; \eta_s), s = 1, 2, \cdots, 2N$ from Eq. (38). Therefore, substituting the obtained $\mu_{-1}(x, t; \widehat{\eta}_s)$ and $\mu_{-2}(x, t; \eta_s), s = 1, 2, \cdots, 2N$ into Eq. (36), and then substituting Eq. (36) into Eq. (35) can yield the $M(x, t; z)$ in terms of the scattering data.

It follows from Eq. (35) with Eq. (36) that the asymptotic behavior of $M(x, t; z)$ is given as

$$M(x,t;z) = I + \frac{M^{(1)}(x,t)}{z} + O\Big(\frac{1}{z^2}\Big), \quad z \to \infty, \tag{40}$$

where

$$M^{(1)}(x,t) = i\sigma_3 Q_- + \sum_{n=1}^{2N}\Big[A_+[\eta_n]\,\mathrm{e}^{-2i\theta(x,t;\eta_n)}\mu_{-2}(x,t;\eta_n),\ A_-[\widehat{\eta}_n]\,\mathrm{e}^{2i\theta(x,t;\widehat{\eta}_n)}\mu_{-1}(x,t;\widehat{\eta}_n)\Big] - \frac{1}{2\pi i}\int_{\Sigma^f} M^+(x,t;\zeta)\,J(x,t;\zeta)\,\mathrm{d}\zeta, \tag{41}$$

with $\mu_{-1}(x,t;\widehat{\eta}_s)$ and $\mu_{-2}(x,t;\eta_s), s = 1, 2, \cdots, 2N$ given by Eqs. (39) and (38).

It follows from Eq. (31) that $M(x,t;z)e^{i\theta(x,t;z)\sigma_3}$ satisfies the Lax pair (4). Substituting $M(x,t;z)e^{i\theta(x,t;z)\sigma_3}$ with Eq. (40) into the x-part of the Lax pair (4) and then choosing the coefficients of z^0 yield the following proposition for the potential $q(x, t)$:

Proposition 3. *The potential with simple poles of the focusing Hirota equation with NZBCs can be expressed as*

$$q(x,t) = q_- - i\sum_{n=1}^{2N} A_-[\widehat{\eta}_n]\,\mathrm{e}^{2i\theta(x,t;\widehat{\eta}_n)}\,\mu_{-11}(x,t;\widehat{\eta}_n) + \frac{1}{2\pi}\int_{\Sigma^f}\big(M^+J\big)_{12}(x,t;\zeta)\,\mathrm{d}\zeta, \tag{42}$$

where $\eta_n = z_n,\ \eta_{N+n} = -\frac{q_0^2}{z_{n-N}^*},\ n = 1, 2, \cdots, N,\ \widehat{\eta}_n = -\frac{q_0^2}{\eta_n}$, *and* $\mu_{-11}(x,t;\widehat{\eta}_n)$*'s are determined by*

$$\sum_{n=1}^{2N}\left(\frac{A_-[\widehat{\eta}_n]\,\mathrm{e}^{2i\theta(x,t;\widehat{\eta}_n)}}{\eta_s - \widehat{\eta}_n} - \frac{iq_-}{\eta_s}\,\delta_{sn}\right)\mu_{-11}(x,t;\widehat{\eta}_n) + \frac{iq_-}{\eta_s} + \int_{\Sigma^f}\frac{(M^+J)_{12}(x,t;\zeta)}{2\pi i(\zeta-\eta_s)}\,\mathrm{d}\zeta = 0,\ s = 1, 2, \ldots, 2N, \tag{43}$$

which can be derived from system (39).

Since $s_{11}(z)$ and $s_{22}(z)$ are analytic in D_+^f and D_-^f, respectively, and the discrete spectral points η_n's and $\widehat{\eta}_n$'s are the simple zeros of $s_{11}(z)$ and $s_{22}(z)$, respectively, then, similarly to Ref. [26] for the focusing NLS equation with NZBCs, one can also find the trace formulae for the focusing Hirota equation with NZBCs

$$s_{11}(z) = e^{s(z)}s_0(z) \text{ for } z \in D_+^f, \quad s_{22}(z) = e^{-s(z)}/s_0(z) \text{ for } z \in D_-^f, \tag{44}$$

where $s(z) = \frac{i}{2\pi}\int_{\Sigma^f}\frac{\log[1+\rho(\zeta)\,\rho^*(\zeta^*)]}{\zeta - z}\,d\zeta$ and $s_0(z) = \prod_{n=1}^{N}\frac{(z-z_n)(z+q_0^2/z_n^*)}{(z-z_n^*)(z+q_0^2/z_n)}$ (see Appendix A or Ref. [26] for the detailed derivation).

According to the limit $z \to 0$ of $s_{11}(z)$ in Eq. (44) and Eq. (30) one can yield the theta condition as

$$\arg(q_+/q_-) = 4\sum_{n=1}^{N}\arg(z_n) + \int_{\Sigma^f}\frac{\log[1+\rho(\zeta)\,\rho^*(\zeta^*)]}{2\pi\zeta}\,d\zeta, \quad z \to 0. \tag{45}$$

Particularly, for the case of the reflectionless potential, that is, $\rho(z) = \hat{\rho}(z) = 0$, which leads to $J = (0)_{2\times 2}$. Thus, Eq. (43) reduces to

$$\sum_{n=1}^{2N}\left(\frac{A_-[\widehat{\eta}_n]\,\mathrm{e}^{2i\theta(x,t;\widehat{\eta}_n)}}{\eta_s - \widehat{\eta}_n} - \frac{iq_-}{\eta_s}\,\delta_{sn}\right)\mu_{-11}(x,t;\widehat{\eta}_n) = -\frac{iq_-}{\eta_s},\ s = 1, 2, \ldots, 2N, \tag{46}$$

which can be solved for $\mu_{-11}(x,t;\widehat{\eta}_n)$ by using the Cramer's rule.

Let $w = \big(w_j\big)_{2N\times 1}$, $v = \big(v_j\big)_{(2N)\times 1}$, $G = \big(g_{sj}\big)_{(2N)\times(2N)}$, and $y = (y_n)_{2N\times 1}$ with $w_j = A_-[\widehat{\eta}_j]\,\mathrm{e}^{2i\theta(x,t;\widehat{\eta}_j)}$, $v_j = -\frac{iq_-}{\widehat{\eta}_j}$, $g_{sj} = \frac{w_j}{\eta_s - \widehat{\eta}_j} + v_s\,\delta_{sj}$, and $y_n = \mu_{-11}(x,t;\widehat{\eta}_n)$. Then it follows from Eq. (46) that $y = (\mu_{-11}(x,t;\widehat{\eta}_n))_{2N\times 1} = G^{-1}v$. And then substituting them into Eq. (42) with $J = 0$ yields the reflectionless potential of the focusing Hirota Eq. (2) in the determinant form:

Proposition 4. *The reflectionless potential with simple poles in the focusing Hirota equation with NZBCs (2) can be found as*

$$q(x,t) = q_- + i\,\frac{\det\begin{bmatrix} G & v \\ w^T & 0\end{bmatrix}}{\det G}. \tag{47}$$

For the case of the reflectionless potential ($\rho(z) = \hat{\rho}(z) = 0$), the above-obtained trace formulae and theta condition reduce to

$$s_{11}(z) = \prod_{n=1}^{N}\frac{(z-z_n)(z+q_0^2/z_n^*)}{(z-z_n^*)(z+q_0^2/z_n)} \text{ for } z \in D_+^f, \quad s_{22}(z) = \prod_{n=1}^{N}\frac{(z-z_n^*)(z+q_0^2/z_n)}{(z-z_n)(z+q_0^2/z_n^*)} \text{ for } z \in D_-^f, \tag{48}$$

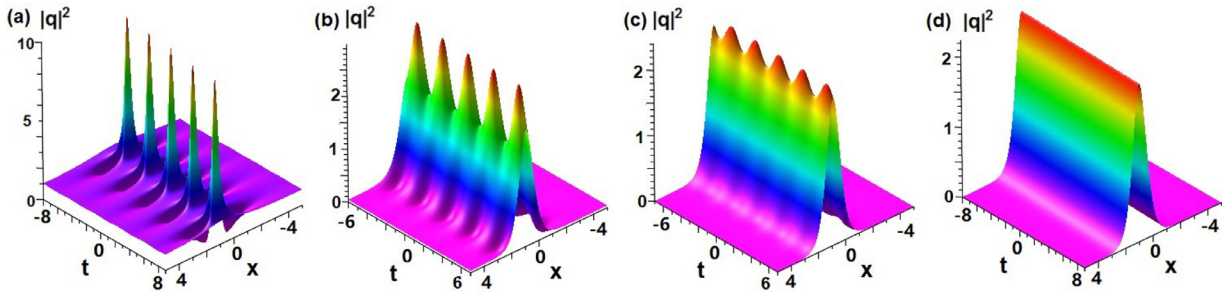


Fig. 2. Simple-pole stationary one-soliton solutions of the focusing Hirota equation with $\alpha = 1$, $\beta = 0.01$ and $N = 1, z_1 = \frac{3}{2}i, A_+[z_1] = 1$. (a): the breather solution with NZBCs and parameters $q_- = 1$; (b) the breather with NZBCs and parameters $q_- = 0.2$; (b) the breather with NZBCs and parameters $q_- = 0.05$; (d) the bright with ZBC and parameter $q_0 \to 0$.

and

$$\arg(q_+/q_-) = 4\sum_{n=1}^{N} \arg(z_n), \tag{49}$$

respectively.

Note that i) when $\alpha = 1$, $\beta = 0$, the simple-pole potential for the focusing Hirota equation with NZBCs in Propositions 3 and 4 can reduce to the known one of the focusing NLS equation with NZBCs [17–20,26]; ii) when $\alpha = 0$, $\beta \neq 0$, the simple-pole potential for the focusing Hirota equation with NZBCs in Propositions 3 and 4 can reduce to one of the focusing complex mKdV equation with NZBCs (3).

Note that for the general parameters α, β, z_n, $A_+[z_n]$ and finite number N, with tha aid of symbolic computation such as *Maple*, we can easily give the explicit expressions for the potential via the formula (47). Generally speaking, these expressions are so complicated for those undetermined parameters. Therefore, in what follows, we investigate the reflectionless potential of the focusing Hirota equation with NZBCs for some known parameters in terms of Proposition 4 with $\alpha\beta \neq 0$:

Case 1. For $N = 1$, $z_1 = 3i/2$, Eq. (49) implies that the asymptotic phase difference is 2π (i.e., there is no phase transition), it follows from Eq. (47) that we have the simple-pole breather solution of the focusing Hirota equation with NZBCs, which is complicated. Here we give a simple example for $\alpha = 1$, $\beta = 0.01$, $q_- = 1$ as

$$q(x,t) = \frac{390\cosh\left(\frac{241}{4320}t - \frac{5}{6}x + \ln\left(\frac{13}{15}\right)\right) - 325\,\mathrm{csch}\left[\frac{1}{2}\ln\left(\frac{81}{16}\right)\right]\sin\left[\frac{65}{36}t - \frac{i}{2}\ln\left(\frac{81}{16}\right)\right]}{390\cosh\left(\frac{241}{4320}t - \frac{5}{6}x + \ln\left(\frac{13}{15}\right)\right) - 360\sin\left(\frac{65}{36}t\right)}, \tag{50}$$

which is a time-periodic breather (also called stationary breather) displayed in Fig. 2(a), and similar to the Kuznetsov-Ma breather for the focusing NLS equation [17,18]. For the fixed $\alpha = 1$, $\beta = 0.01$, we choose other NZBCs such as $q_- = 0.2$, 0.05 to find the corresponding simple-pole breather solutions of the focusing Hirota equation (see Fig. 2(b) and (c)). It follows from Fig. 2(a)-(c) that when the non-zero parameter q_- becomes smaller, the periodic behavior of the simple-pole breather only appears in its top part, and the maximal amplitude under the background gradually decreases. In particular, when $q_0 \to 0$, the breather solution of the focusing Hirota equation with NZBCs becomes the simple-pole bright soliton of the focusing Hirota equation with ZBC (see Fig. 2(d)).

Case 2. For $N = 1$, $z_1 = ae^{i\nu}$ with $a > 1$ and $\nu \in (0, \pi/2) \cup (\pi/2, \pi)$, the asymptotic phase difference is $\arg(q_+/q_-) = 4\nu \in (0, 2\pi)$. The focusing Hirota equation with NZBCs possesses the non-stationary solitons. In particular, when $\alpha = 1$, $\beta = 0$, these non-stationary solitons reduce to ones of the focusing NLS equation with NZBCs [19,20]. For example, for $N = 1$, $\alpha = 1$, $\beta = 0.01$, $z_1 = 2e^{i\pi/4}$, $A_+[z_1] = 1$, we have the non-stationary soliton of the focusing Hirota equation with NZBCs as

$$q(x,t) = -\frac{40\sqrt{6}\cosh[\xi_2 - \ln(25/136)/2] + 30\sqrt{17}i\sin[\xi_1 - \arctan(4) + 2\ln(2)i]}{40\sqrt{6}\cosh[\xi_2 + \ln(25/136)/2] + 16\sqrt{17}\cos(\xi_1 + \arccos(4/\sqrt{17}))}, \tag{51}$$

where $\xi_1 = 5\sqrt{2}(t - 16x)/64$, $\xi_2 = 3\sqrt{2}x/4 + (27\sqrt{2}/1600 - 17/4)t$, and the asymptotic phase difference is π (see Fig. 3(a) and (b)). If we only change $z_1 = 2e^{3i\pi/4}$, then we have another non-stationary soliton of the Hirota equation with NZBCs (see Fig. 3(c) and (d)), whose asymptotic phase difference is 3π, this is, the phase difference is also π. But the slopes of motions of the hump centers in every period are opposite (see Fig. 3(b) and (d)).

Case 3. For $N = 1$, $q_- = 1$, $z_1 \to e^{i\pi/4}$, $\alpha = 1$, $\beta = 0.01$, we have the non-stationary breather of the focusing Hirota equation with NZBCs

$$q(x,t) = \frac{2\sinh(2t) - \mathrm{csch}\left(\frac{\ln(3)}{2}\right)\sin\left(\frac{\sqrt{2}}{25}t - x - \frac{i}{2}\ln(3)\right)}{2\cosh(2t) + \sqrt{2}\cos\left(\frac{\sqrt{2}}{25}t - x + \frac{\pi}{4}\right)}, \tag{52}$$

which is a space-periodic breather (also called non-stationary breather) displayed in Fig. 4(a) and (b), and similar to the Akhmediev breather for the focusing NLS equation [21]. Fig. 4(b) implies that the motion of the peak centers in each period

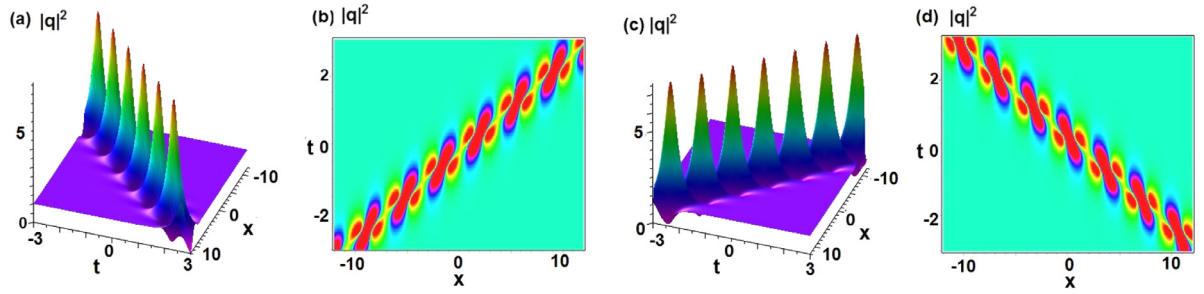


Fig. 3. Simple-pole non-stationary breather solutions of the focusing Hirota equation with NZBCs and $\alpha = 1$, $\beta = 0.01$ and $N = 1$, $q_- = 1$, $A_+[z_1] = 1$. (a) $z_1 = 2e^{i\pi/4}$; (b) $z_1 = 2e^{3\pi i/4}$.

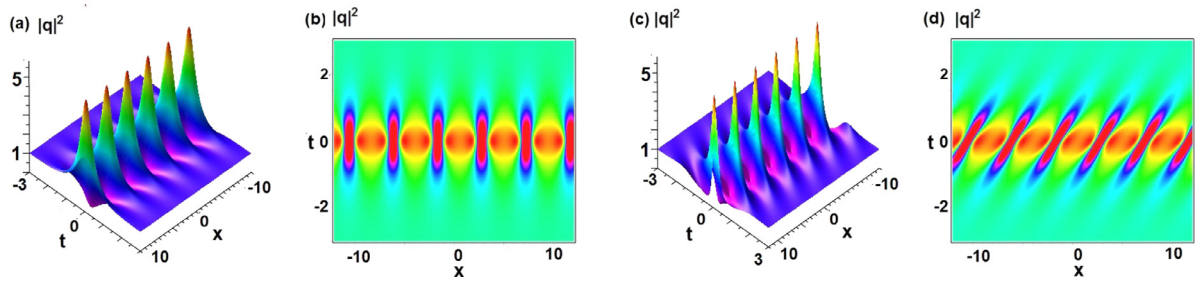


Fig. 4. Simple-pole non-stationary AKmediev breathers of the focusing Hirota equation with NZBCs and $N = 1$, $\alpha = 1$, $q_- = 1$, $z_1 \to e^{\pi i/4}$, $A_+[z_1] = 1$. (a) $\beta = 0.01$; (b) $\beta = 0.5$.

is almost vertical to the x-axis for the small third-order dispersive parameter $\beta = 0.01$. If we only increase β as $\beta = 0.5$, then we have the modified Akhmediev breather (see Fig. 4(c) and (d))

$$q(x,t) = \frac{2\sinh(2t) - i\sqrt{2}\cos\left(2t - x + \frac{\pi}{4}\right)}{2\cosh(2t) + \sqrt{2}\cos\left(2t - x + \frac{\pi}{4}\right)}, \tag{53}$$

where the motion of hump centers in each period is not vertical to the x-axis (see Fig. 4(d)).

Case 4. For $N = 2$, we have the weak interactions of simple-pole breather-breather solutions of the focusing Hirota equation with NZBCs (see Fig. 5(a1)-(a3) for $q_- = 1$ and Fig. 5(b1)-(b3) for $q_- = 0.2$, where $z_1 = 1/5 + 3/2i$, $z_2 = -1/5 + 3/2i$). It follows from Fig. 5(b1)-(b3) that when the parameter q_- becomes smaller (e.g., $q_- = 0.2$), the periodic behaviors of the simple-pole breather-breather solutions only appear in their top parts. Particularly, when $q_0 \to 0$, we have the weak interactions of the simple-pole bright-bright solitons of the focusing Hirota equation with ZBC (see Fig. 5(c1)-(c3)).

Case 5. For $N = 2$, if we take two distinct discrete eigenvalues as $z_1 = \frac{1}{5} + 2i$, $z_2 = 1 + i$, then we have the strong interactions of simple-pole breather-breather solutions of the focusing Hirota equation with NZBCs such that the larger amplitude can be generated when two breathers interact (see Fig. 6(a1)-(a3) for $q_- = 1$ and Fig. 6(b1)-(b3) for $q_- = 0.3$). Particularly, when $q_0 \to 0$, we have the strong interactions of the simple-pole bright-bright solitons of the focusing Hirota equation with ZBC (see Fig. 6(c1)-(c3)). If we take z_1 and z_2 with the same real parts and almost same imaginary parts, e.g., $z_1 = -1/65 + 3i/2$, $z_1 = -1/65 + 31i/20$, then we have another type of simple-pole breather-breather solutions (see Fig. 7(a1)-(a3) for $q_- = 1$ and Fig. 7(b1)-(b3) for $q_- = 0.2$). Particularly, when $q_0 \to 0$, we have the simple-pole bright-bright solitons of the focusing Hirota equation with ZBC (see Fig. 7(c1)-(c3)).

Similarly, for $N > 2$, we also have the higher-order simple-pole breather and bright soliton solutions, which are omitted here. Note that these simple-pole breather and bright solitons solutions of the focusing Hirota equation could be derived via the Darboux transformation [38].

2.3. The focusing hirota equation with NZBCs: Double poles

The direct and inverse scattering problems in the case of double poles are similar to ones in the case of simple poles, but they also have some differences, that is, they can make the matrix function $M(x, t; z)$ in the Riemann-Hilbert problem exhibit distinct types of poles (i.e., simple poles and double poles) such that the function $M(x, t; z)$ can further generate distinct potentials for the simple poles and double poles. Moreover, the trace formulae and theta conditions are also different for the two cases (see [30] for the focusing NLS equation with double poles).

In the following, we suppose that the discrete spectral points Z^f are double zeros of the scattering coefficients $s_{11}(z)$ and $s_{22}(z)$, that is, $s_{11}(z_0) = s'_{11}(z_0) = 0$, $s''_{11}(z_0) \neq 0$ for $\forall z_0 \in Z^f \cap D^f_+$, and $s_{22}(z_0) = s'_{22}(z_0) = 0$, $s''_{22}(z_0) \neq 0$ for $\forall z_0 \in Z^f \cap D^f_-$. For convenience, we give a simple proposition [30]: if $f(z)$ and $g(z)$ are analytic in some complex domain Ω, and $z_0 \in \Omega$ is

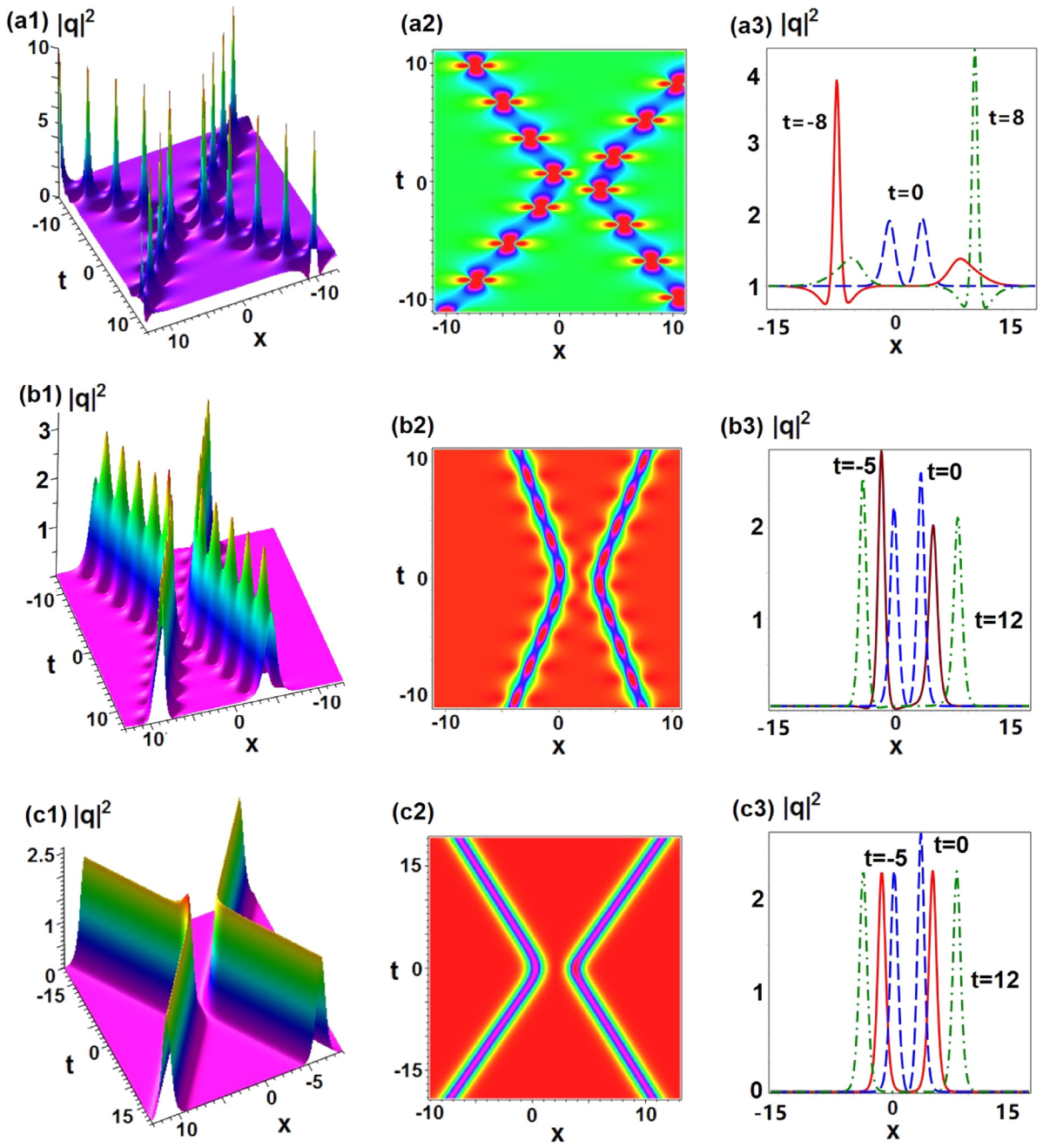


Fig. 5. Weak interactions of simple-pole two-soliton solutions of the focusing Hirota equation with $\alpha = 1$, $\beta = 0.01$, $N = 2$, $z_1 = \frac{1}{5} + \frac{3}{2}i$, $z_2 = -\frac{1}{5} + \frac{3}{2}i$, $A_+[z_1] = A_+[z_2] = 1$. (a1, a2, a3): the breather-breather solutions with NZBCs and parameter $q_- = 1$; (b1, b2, b3): the breather-breather solutions with NZBCs and parameter $q_- = 0.2$; (c1, c2, c3): the bright-bright solitons with ZBC and parameter $q_- = q_0 \to 0$.

a double zero of $g(z)$ and $f(z_0) \neq 0$, then the function $f(z)/g(z)$ has the double pole $z = z_0$, and its residue Res$[f/g]$ and the coefficient $P_{-2}[f/g]$ of $(z - z_0)^{-2}$ in the Laurent series are given by

$$\underset{z=z_0}{\mathrm{P}_{-2}}[f/g] = \frac{2f(z_0)}{g''(z_0)}, \qquad \underset{z=z_0}{\mathrm{Res}}[f/g] = 2\left(\frac{f'(z_0)}{g''(z_0)} - \frac{f(z_0)g'''(z_0)}{3[g''(z_0)]^2}\right). \tag{54}$$

Similarly to [30] for the focusing NLS equation with double poles, for $s_{11}(z_0) = s'_{11}(z_0) = 0$, $s''_{11}(z_0) \neq 0$ in $\forall z_0 \in Z^f \cap D^f_+$, Eq. (22) still holds (the coefficient $b_+(z_0)$ may be distinct, we here still use the same sign). The first one of Eq. (16) can be

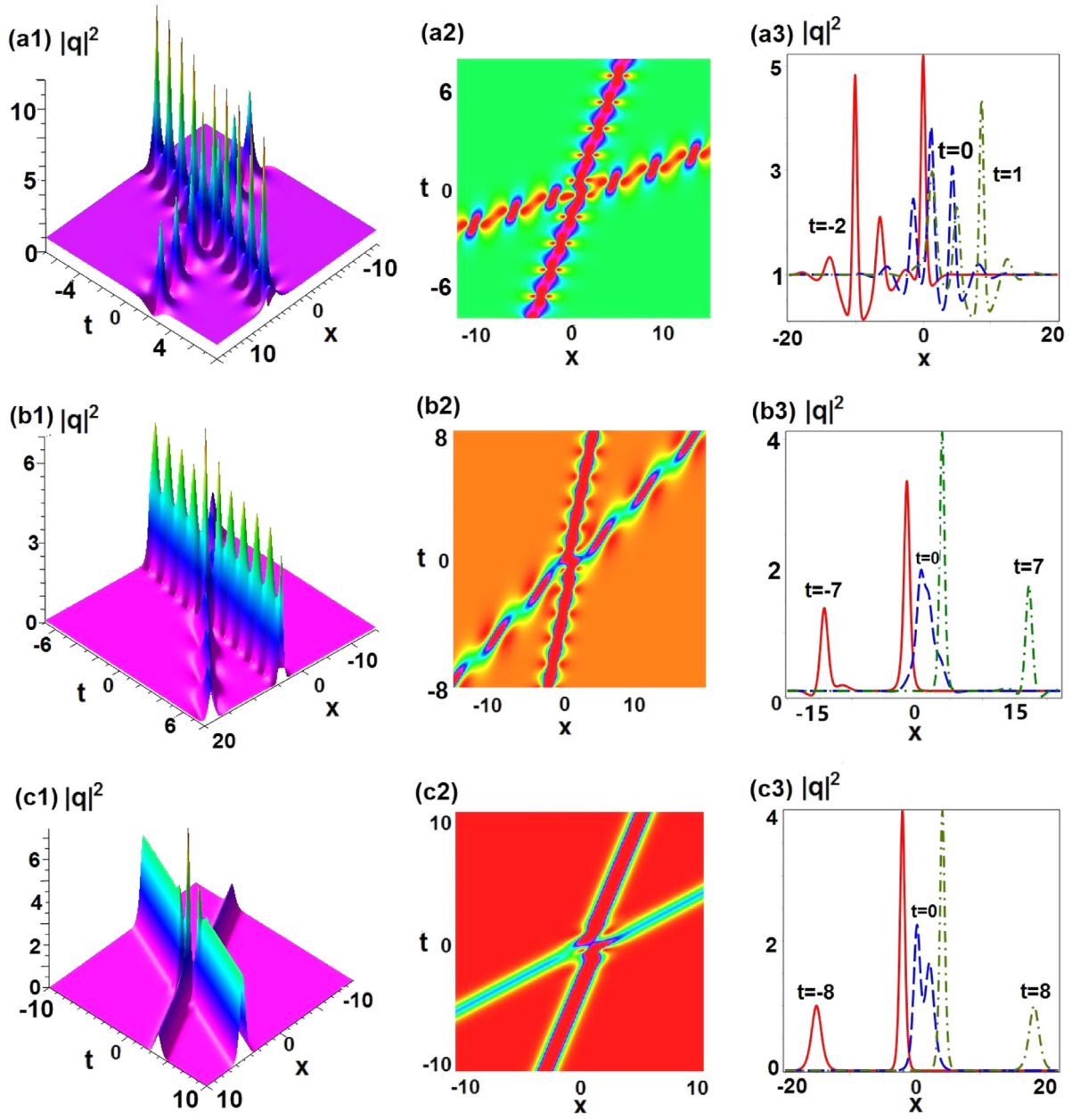


Fig. 6. Strong interactions of simple-pole two-soliton solutions of the focusing Hirota equation with $\alpha = 1$, $\beta = 0.01$, $N = 2$, $z_1 = 1/5 + 2i$, $z_2 = 1 + i$, $A_+[z_1] = A_+[z_2] = 1$. (a1, a2, a3): the breather-breather solutions with NZBCs and parameter $q_- = 1$; (b1, b2, b3): the breather-breather solutions with NZBCs and parameter $q_- = 0.3$; (c1, c2, c3): the bright-bright solitons with ZBC and parameter $q_0 \to 0$.

rewritten as

$$s_{11}(z)\gamma_f(z) = |\Phi_{+1}(x,t;z),\ \Phi_{-2}(x,t;z)|, \tag{55}$$

whose first-order partial derivative with respect to z yields

$$[s_{11}(z)\gamma_f(z)]' = |\Phi'_{+1}(x,t;z),\ \Phi_{-2}(x,t;z)| + |\Phi_{+1}(x,t;z),\ \Phi'_{-2}(x,t;z)|. \tag{56}$$

Taking $z = z_0 \in Z^f \cap D^f_+$ in Eq. (56) and using $s_{11}(z_0) = s'_{11}(z_0) = 0$ and Eq. (22) yield

$$|\Phi'_{+1}(x,t;z_0) - b_+(z_0)\Phi'_{-2}(x,t;z_0),\ \Phi_{-2}(x,t;z_0)| = 0, \tag{57}$$

which implies that there exists another constant $c_+(z_0)$ such that

$$\Phi'_{+1}(x,t;z_0) = c_+(z_0)\Phi_{-2}(x,t;z_0) + b_+(z_0)\Phi'_{-2}(x,t;z_0). \tag{58}$$

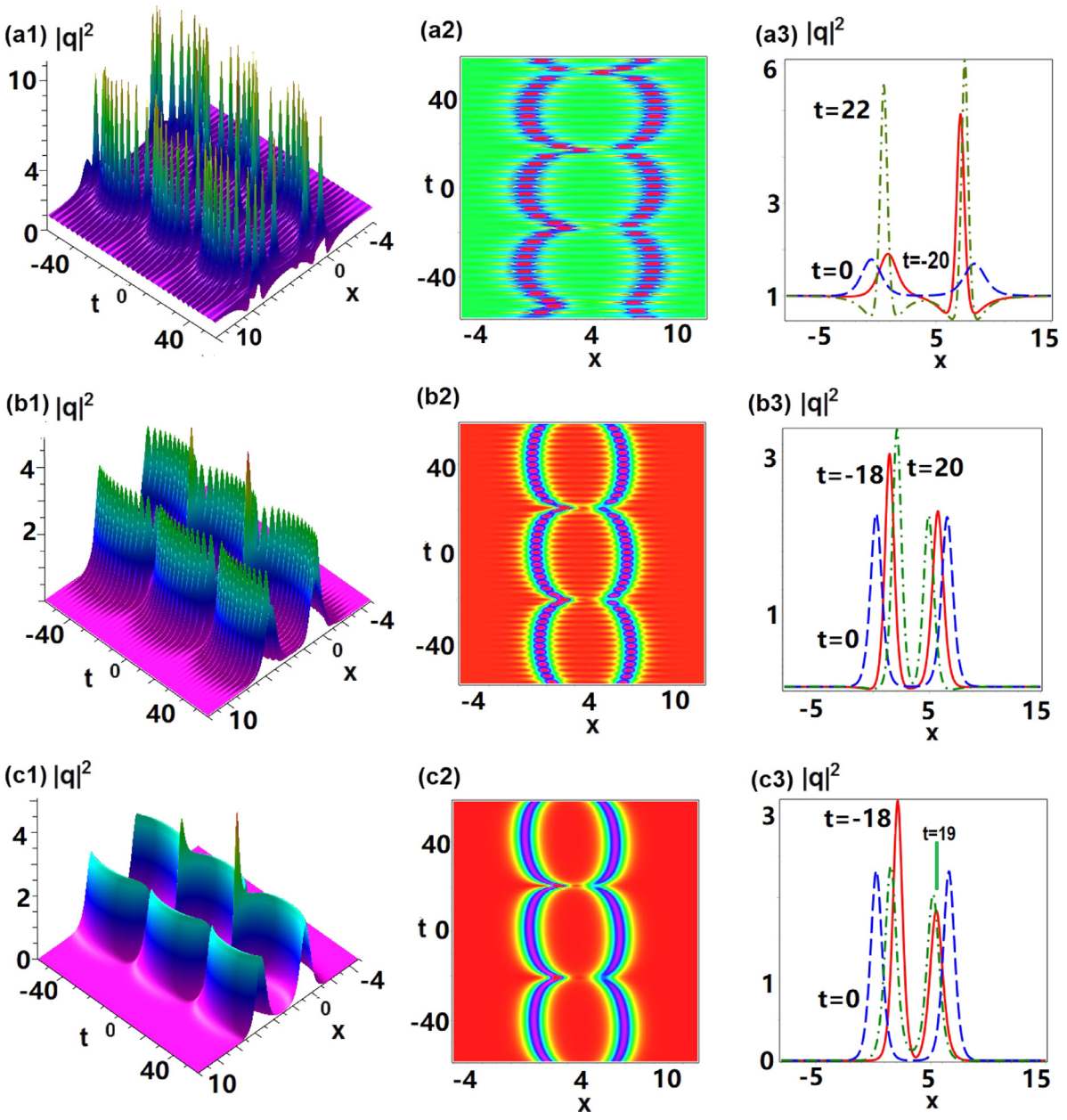


Fig. 7. Strong interactions of simple-pole two-soliton solutions of the focusing Hirota equation with $\alpha = 1$, $\beta = 0.01$, $N = 2$, $z_1 = -1/65 + 3i/2$, $z_2 = -1/65 + 31i/20$, $A_+[z_1] = A_+[z_2] = 1$. (a1, a2, a3): the breather-breather solutions with NZBCs and parameter $q_- = 1$; (b1, b2, b3): the breather-breather solutions with NZBCs and parameter $q_- = 0.2$; (c1, c2, c3): the bright-bright solitons with ZBC and parameter $q_0 \to 0$.

According to Eqs. (22), (54) and (58), we have

$$\begin{aligned}
&\operatorname*{P}_{z=z_0}\left[\frac{\Phi_{+1}(x,t;z)}{s_{11}(z)}\right] = \frac{2\Phi_{+1}(x,t;z_0)}{s''_{11}(z_0)} = \frac{2b_+(z_0)}{s''_{11}(z_0)}\Phi_{-2}(x,t;z_0) = A_+[z_0]\Phi_{-2}(x,t;z_0),\\
&\operatorname*{Res}_{z=z_0}\left[\frac{\Phi_{+1}(x,t;z)}{s_{11}(z)}\right] = \frac{2\Phi'_{+1}(x,t;z_0)}{s''_{11}(z_0)} - \frac{2\Phi_{+1}(x,t;z_0)s'''_{11}(z_0)}{3(s''_{11}(z_0))^2}\\
&\qquad = \frac{2b_+(z_0)}{s''_{11}(z_0)}\left[\Phi'_{-2}(x,t;z_0) + \left(\frac{c_+(z_0)}{b_+(z_0)} - \frac{s'''_{11}(z_0)}{3s''_{11}(z_0)}\right)\Phi_{-2}(x,t;z_0)\right]\\
&\qquad = A_+[z_0]\left[\Phi'_{-2}(x,t;z_0) + B_+[z_0]\Phi_{-2}(x,t;z_0)\right].
\end{aligned} \tag{59}$$

Similarly, for $s_{22}(z_0^*)=s_{22}'(z_0^*)=0,\ s_{22}''(z_0^*)\neq 0$ in $\forall z_0^*\in Z^f\cap D_-^f$, Eq. (24) still holds (the coefficient $b_-(z_0^*)$ may be distinct, we here still use the same sign). It follows from the second one of Eqs. (16) and (24) that

$$\Phi_{+2}'(x,t;z_0^*)=c_-(z_0^*)\Phi_{-1}(x,t;z_0^*)+b_-(z_0^*)\Phi_{-1}'(x,t;z_0^*) \tag{60}$$

for some constant $c_-(z_0^*)$.

According to Eqs. (24), (54), and (60), we have

$$\begin{aligned}
&\mathop{\mathrm{P}_{-2}}_{z=z_0^*}\left[\frac{\Phi_{+2}(x,t;z)}{s_{22}(z)}\right]=\frac{2\Phi_{+2}(x,t;z_0^*)}{s_{22}''(z_0^*)}=\frac{2b_-(z_0^*)}{s_{22}''(z_0^*)}\Phi_{-1}(x,t;z_0^*)=A_-[z_0^*]\Phi_{-1}(x,t;z_0^*),\\
&\mathop{\mathrm{Res}}_{z=z_0^*}\left[\frac{\Phi_{+2}(x,t;z)}{s_{22}(z)}\right]=\frac{2\Phi_{+2}'(x,t;z_0^*)}{s_{22}''(z_0^*)}-\frac{2\Phi_{+2}(x,t;z_0^*)s_{22}'''(z_0^*)}{3(s_{22}''(z_0^*))^2}\\
&\qquad=\frac{2b_-(z_0^*)}{s_{22}''(z_0^*)}\left[\Phi_{-1}'(x,t;z_0^*)+\left(\frac{c_-(z_0^*)}{b_-(z_0^*)}-\frac{s_{22}'''(z_0^*)}{3s_{22}''(z_0^*)}\right)\Phi_{-2}(x,t;z_0^*)\right]\\
&\qquad=A_-[z_0^*]\left[\Phi_{-1}'(x,t;z_0^*)+B_-[z_0^*]\Phi_{-1}(x,t;z_0^*)\right].
\end{aligned} \tag{61}$$

Therefore, we know

$$\begin{aligned}
&A_+[z_0]=\frac{2\,b_+[z_0]}{s_{11}''(z_0)},\quad B_+[z_0]=\frac{c_+(z_0)}{b_+(z_0)}-\frac{s_{11}'''(z_0)}{3s_{11}''(z_0)},\quad z_0\in Z^f\cap D_+^f,\\
&A_-[z_0^*]=\frac{2\,b_-[z_0^*]}{s_{22}''(z_0^*)},\quad B_-[z_0^*]=\frac{c_-(z_0^*)}{b_-(z_0^*)}-\frac{s_{22}'''(z_0^*)}{3s_{22}''(z_0^*)},\quad z_0^*\in Z^f\cap D_-^f,
\end{aligned}$$

from which we have the symmetries

$$A_+[z_0]=-A_-^*[z_0^*]=\frac{z_0^4q_-^*}{q_0^4q_-}A_-[-q_0^2/z_0],\quad B_+[z_0]=B_-^*[z_0^*]=\frac{q_0^2}{z_0^2}B_-[-q_0^2/z_0]+2/z_0,\quad z_0\in Z^f\cap D_+^f, \tag{62}$$

which lead to

$$\begin{aligned}
&A_+[z_n]=-A_-^*[z_n^*]=\frac{z_n^4q_-^*}{q_0^4q_-}A_-[-q_0^2/z_n]=-\frac{z_n^4q_-^*}{q_0^4q_-}A_+[-q_0^2/z_n^*],\quad z_n\in Z^f\cap D_+^f,\\
&B_+[z_n]=B_-^*[z_n^*]=\frac{q_0^2}{z_n^2}B_-[-q_0^2/z_n]+2/z_n=\frac{q_0^2}{z_n^2}B_+^*[-q_0^2/z_n^*]+2/z_n,\quad z_n\in Z^f\cap D_+^f.
\end{aligned} \tag{63}$$

It follows from Eqs. (12), (32), (59), and (61) that

$$\begin{aligned}
&\mathop{\mathrm{P}_{-2}}_{z=\eta_n}M_1^+(x,t;z)=\mathop{\mathrm{P}_{-2}}_{z=\eta_n}\left[\frac{\mu_{+1}(x,t;z)}{s_{11}(z)}\right]=A[\eta_n]\,\mathrm{e}^{-2i\theta(x,t;\eta_n)}\mu_{-2}(x,t;\eta_n),\\
&\mathop{\mathrm{P}_{-2}}_{z=\widehat{\eta}_n}M_2^-(x,t;z)=\mathop{\mathrm{P}_{-2}}_{z=\widehat{\eta}_n}\left[\frac{\mu_{+2}(x,t;z)}{s_{22}(z)}\right]=A[\widehat{\eta}_n]\,\mathrm{e}^{2i\theta(x,t;\widehat{\eta}_n)}\mu_{-1}(x,t;\widehat{\eta}_n),\\
&\mathop{\mathrm{Res}}_{z=\eta_n}M_1^+(x,t;z)=\mathop{\mathrm{Res}}_{z=\eta_n}\left[\frac{\mu_{+1}(x,t;z)}{s_{11}(z)}\right]=A[\eta_n]\,\mathrm{e}^{-2i\theta(x,t;\eta_n)}\{\mu_{-2}'(x,t;\eta_n)+[B[\eta_n]-2\,i\,\theta'(x,t;\eta_n)]\mu_{-2}(x,t;\eta_n)\},\\
&\mathop{\mathrm{Res}}_{z=\widehat{\eta}_n}M_2^-(x,t;z)=\mathop{\mathrm{Res}}_{z=\widehat{\eta}_n}\left[\frac{\mu_{+2}(x,t;z)}{s_{22}(z)}\right]=A[\widehat{\eta}_n]\,\mathrm{e}^{2i\theta(x,t;\widehat{\eta}_n)}\{\mu_{-1}'(x,t;\widehat{\eta}_n)+[B[\widehat{\eta}_n]+2\,i\,\theta'(x,t;\widehat{\eta}_n)]\mu_{-1}(x,t;\widehat{\eta}_n)\}.
\end{aligned} \tag{64}$$

The Riemann-Hilbert problem in Proposition 2 still holds for the case of double poles. To solve this kind of Riemann-Hilbert problem, one has to subtract out the asymptotic values as $z\to\infty$ and $z\to 0$ and the singularity contributions

$$M_{dp}(x,t;z)=I+\frac{i}{z}\sigma_3\,Q_-+\sum_{n=1}^{2N}M_{dp}^{(n)},\quad M_{dp}^{(n)}=\frac{\mathop{\mathrm{P}_{-2}}_{z=\eta_n}M^+}{(z-\eta_n)^2}+\frac{\mathop{\mathrm{Res}}_{z=\eta_n}M^+}{z-\eta_n}+\frac{\mathop{\mathrm{P}_{-2}}_{z=\widehat{\eta}_n}M^-}{(z-\widehat{\eta}_n)^2}+\frac{\mathop{\mathrm{Res}}_{z=\widehat{\eta}_n}M^-}{z-\widehat{\eta}_n} \tag{65}$$

from the jump condition $M^-=M^+(I-J)$ such that we have

$$M^-(x,t;z)-M_{dp}(x,t;z)=M^+(x,t;z)-M_{dp}(x,t;z)-M^+(x,t;z)\,J, \tag{66}$$

in which, $M^\pm(x,t;z)-M_{dp}(x,t;z)$ are analytic in $D_\pm^f$. Moreover, their asymptotics are both $O\left(\frac{1}{z}\right)$ as $z\to\infty$ and $O(1)$ as $z\to 0$, and $J(x,t;z)$ is $O\left(\frac{1}{z}\right)$ as $z\to\infty$, and $O(z)$ as $z\to 0$. Therefore, the Cauchy projectors and Plemelj's formulae are used to solve Eq. (66) to yield

$$M(x,t;z)=M_{dp}(x,t;z)+\frac{1}{2\pi i}\int_{\Sigma^f}\frac{M^+(x,t;\zeta)J(x,t;\zeta)}{\zeta-z}\,\mathrm{d}\zeta,\quad z\in\mathbb{C}\backslash\Sigma^f, \tag{67}$$

where $\int_{\Sigma^f}$ stands for the integral along the oriented contours shown in Fig. 1(right).

Therefore, by means of Eq. (64), the parts of $\mathrm{P}_{-2}(\cdot)$ and $\mathrm{Res}\,(\cdot)$ in Eq. (67) can be found as

$$M_{dp}^{(n)} = \left(C_n(z)\left[\mu'_{-2}(\eta_n) + \left(D_n + \frac{1}{z-\eta_n}\right)\mu_{-2}(\eta_n)\right], \widehat{C}_n(z)\left[\mu'_{-1}(\widehat{\eta}_n) + \left(\widehat{D}_n + \frac{1}{z-\widehat{\eta}_n}\right)\mu_{-1}(\widehat{\eta}_n)\right]\right), \tag{68}$$

where

$$C_n(z) = \frac{A_+[\eta_n]}{z-\eta_n}\,\mathrm{e}^{-2i\theta(\eta_n)},\ D_n = B_+[\eta_n] - 2\,i\,\theta'(\eta_n),\ \widehat{C}_n(z) = \frac{A_-[\widehat{\eta}_n]}{z-\widehat{\eta}_n}\,\mathrm{e}^{2i\theta(\widehat{\eta}_n)},\ \widehat{D}_n = B_-[\widehat{\eta}_n] + 2\,i\,\theta'(\widehat{\eta}_n). \tag{69}$$

To determine $\mu'_{-2}(\eta_n), \mu_{-2}(\eta_n), \mu'_{-1}(\eta_n)$, and $\mu_{-1}(\eta_n)$ in Eq. (68), for $z = \eta_s$ $(s = 1, 2, \cdots, 2N)$, it follows from the second column of $M(x, t; \lambda)$ given by Eq. (67) with Eq. (68) that

$$\mu_{-2}(z) = \begin{bmatrix} \frac{iq_-}{z} \\ 1 \end{bmatrix} + \sum_{n=1}^{2N} \widehat{C}_n(z)\left[\mu'_{-1}(\widehat{\eta}_n) + \left(\widehat{D}_n + \frac{1}{z-\widehat{\eta}_n}\right)\mu_{-1}(\widehat{\eta}_n)\right] + \frac{1}{2\pi i}\int_{\Sigma^f} \frac{(M^+J)_2(\zeta)}{\zeta - z}\,\mathrm{d}\zeta. \tag{70}$$

whose first-order derivative with respect to z yields

$$\mu'_{-2}(z) = \begin{bmatrix} -\frac{iq_-}{z^2} \\ 0 \end{bmatrix} - \sum_{n=1}^{2N} \frac{\widehat{C}_n(z)}{z-\widehat{\eta}_n}\left[\mu'_{-1}(\widehat{\eta}_n) + \left(\widehat{D}_n + \frac{2}{z-\widehat{\eta}_n}\right)\mu_{-1}(\widehat{\eta}_n)\right] + \frac{1}{2\pi i}\int_{\Sigma^f} \frac{(M^+J)_2(\zeta)}{(\zeta - z)^2}\,\mathrm{d}\zeta. \tag{71}$$

Moreover, it follows from Eq. (18) that

$$\mu'_{-2}(z) = -\frac{iq_-}{z^2}\,\mu_{-1}(-q_0^2/z) + \frac{iq_0^2 q_-}{z^3}\,\mu'_{-1}(-q_0^2/z). \tag{72}$$

The substitution of Eqs. (38) and (72) into Eqs. (70) and (71) with $z = \eta_s, s = 1, 2, \cdots, 2N$ yields an algebraic system of $4N$ equations with the $4N$ unknowns $\mu_{-1}(\widehat{\eta}_n), \mu'_{-1}(\widehat{\eta}_n), n = 1, 2, \cdots, 2N$ in the form

$$\begin{aligned}
&\sum_{n=1}^{2N} \widehat{C}_n(\eta_s)\,\mu'_{-1}(\widehat{\eta}_n) + \left[\widehat{C}_n(\eta_k)\left(\widehat{D}_n + \frac{1}{\eta_s - \widehat{\eta}_n}\right) - \frac{iq_-}{\eta_s}\,\delta_{sn}\right]\mu_{-1}(\widehat{\eta}_n) = -\begin{bmatrix} \frac{iq_-}{\eta_s} \\ 1 \end{bmatrix} - \frac{1}{2\pi i}\int_{\Sigma^f} \frac{(M^+J)_2(\zeta)}{\zeta - \eta_k}\,\mathrm{d}\zeta, \\
&\sum_{n=1}^{2N}\left(\frac{\widehat{C}_n(\eta_s)}{\eta_s - \widehat{\eta}_n} + \frac{iq_0^2 q_-}{\eta_s^3}\delta_{sn}\right)\mu'_{-1}(\widehat{\eta}_n) + \left[\frac{\widehat{C}_n(\eta_s)}{\eta_s - \widehat{\eta}_n}\left(\widehat{D}_n + \frac{2}{\eta_s - \widehat{\eta}_n}\right) - \frac{iq_-}{\eta_s^2}\delta_{sn}\right]\mu_{-1}(\widehat{\eta}_n) = \begin{bmatrix} -\frac{iq_-}{\eta_s^2} \\ 0 \end{bmatrix} + \int_{\Sigma^f} \frac{(M^+J)_2(\zeta)}{2\pi i(\zeta - \eta_k)^2}\,\mathrm{d}\zeta.
\end{aligned} \tag{73}$$

which can generate $\mu_{-1}(x, t; \widehat{\eta}_n), \mu'_{-1}(x, t; \widehat{\eta}_n), n = 1, 2, \cdots, 2N$ such that one can also get $\mu_{-2}(x, t; \eta_n)$, $\mu'_{-2}(x, t; \eta_n)$, $n = 1, 2, \cdots, 2N$ from Eqs. (38) and (72). Therefore, substituting them into Eq. (68), and then substituting Eq. (68) into Eq. (67) can yield the $M(x, t; z)$ in terms of the scattering data.

It follows from Eq. (67) with Eq. (68) that the asymptotic behavior of $M(x, t; z)$ is still of the form (40), but $M^{(1)}(x, t)$ is here replaced by

$$\begin{aligned}
M^{(1)}(x,t) = {} & i\sigma_3\,\mathrm{Q}_- - \frac{1}{2\pi i}\int_{\Sigma^f} \left(M^+J\right)(\zeta)\,d\zeta \\
& + \sum_{n=1}^{2N}\left[A_+[\eta_n]\,\mathrm{e}^{-2i\theta(\eta_n)}(\mu'_{-2}(\eta_n) + D_n\,\mu_{-2}(\widehat{\eta}_n)),\ A_-[\widehat{\eta}_n]\,\mathrm{e}^{2i\theta(\widehat{\eta}_n)}(\mu'_{-1}(\widehat{\eta}_n) + \widehat{D}_n\,\mu_{-1}(\widehat{\eta}_n))\right].
\end{aligned} \tag{74}$$

It follows from Eq. (31) that $M(x, t; z)e^{i\theta(x,t;z)\sigma_3}$ satisfies the Lax pair (4). Substituting $M(x, t; z)e^{i\theta(x,t;z)\sigma_3}$ with Eq. (40) and (74) into the x-part of the Lax pair (4) and then choosing the coefficients of z^0 yield the following property for the potential $q(x, t)$ for the case of double poles:

Proposition 5. *The potential with double poles of the focusing Hirota equation with NZBCs is given by*

$$q(x,t) = q_- - i\sum_{n=1}^{2N} A_-[\widehat{\eta}_n]\,\mathrm{e}^{2i\theta(\widehat{\eta}_n)}\left(\mu'_{-11}(\widehat{\eta}_n) + \widehat{D}_n\,\mu_{-11}(\widehat{\eta}_n)\right) + \frac{1}{2\pi}\int_{\Sigma^f}\left(M^+J\right)_{12}(\zeta)\,\mathrm{d}\zeta, \tag{75}$$

where $\widehat{C}_n(z) = \frac{A[\widehat{\eta}_n]}{z-\widehat{\eta}_n}\,\mathrm{e}^{2i\theta(\widehat{\eta}_n)}$, $\widehat{D}_n = B[\widehat{\eta}_n] + 2\,i\,\theta'(\widehat{\eta}_n)$, *and* $\mu'_{-11}(\widehat{\eta}_n)$ *and* $\mu_{-11}(\widehat{\eta}_n)$, $n = 1, 2, \cdots, 2N$ *are given by Eq. (73).*

Similarly to the case of simple poles, we also have the trace formulae for the case of double poles as

$$s_{11}(z) = e^{s(z)}s_0(z)\ \text{ for }\ z \in D_+^f, \quad s_{22}(z) = e^{-s(z)}/s_0(z)\ \text{ for }\ z \in D_-^f, \tag{76}$$

where $s(z) = \frac{i}{2\pi}\int_{\Sigma^f} \frac{\log[1+\rho(\zeta)\,\rho^*(\zeta^*)]}{\zeta - z}\,d\zeta$ and $s_0(z) = \prod_{n=1}^{N} \frac{(z-z_n)^2(z+q_0^2/z_n^*)^2}{(z-z_n^*)^2(z+q_0^2/z_n)^2}$. They are the same ones as those for the focusing NLS equation with double poles [30].

According to the limit $z \to 0$ of $s_{11}(z)$ in Eq. (76) and Eq. (30), one can yield the theta condition as

$$\arg\left(q_+/q_-\right) = 8\sum_{n=1}^{N}\arg(z_n) + \int_{\Sigma^f} \frac{\log[1 + \rho(\zeta)\,\rho^*(\zeta^*)]}{2\pi\,\zeta}\,\mathrm{d}\zeta, \quad z \to 0. \tag{77}$$

In particular, for the reflectionless case, that is, $\rho(z)=\hat{\rho}(z)=0$, we have the following proposition:

Proposition 6. *The reflectionless potential with double poles of the focusing Hirota equation with NZBCs is given as*

$$q(x,t)=q_{-}+i\frac{\det\begin{bmatrix}H & v\\ w^{T} & 0\end{bmatrix}}{\det H}, \tag{78}$$

where $H=(H^{(sj)})_{2\times 2}$, $H^{(sj)}=(h_{kn}^{(sj)})_{(2N)\times(2N)}$, $h_{kn}^{(11)}=\widehat{C}_n(\eta_k)(\widehat{D}_n+\frac{1}{\eta_k-\widehat{\eta}_n})-\frac{iq_{-}}{\eta_k}\delta_{k,n}$, $h_{kn}^{(12)}=\widehat{C}_n(\eta_k)$, $h_{kn}^{(21)}=\frac{\widehat{C}_n(\eta_k)}{\eta_k-\widehat{\eta}_n}(\widehat{D}_n+\frac{2}{\eta_k-\widehat{\eta}_n})-\frac{iq_{-}}{\eta_k^2}\delta_{kn}$, $h_{kn}^{(22)}=\frac{\widehat{C}_n(\eta_k)}{\eta_k-\widehat{\eta}_n}+\frac{iq_{-}q_0^2}{\eta_k^3}\delta_{kn}$, $y=(y_1^{(1)},y_2^{(1)},\cdots,y_{2N}^{(1)},y_1^{(2)},y_2^{(2)},\cdots,y_{2N}^{(2)})^{T}$, $y=w,v$, $w_n^{(1)}=A_{-}[\widehat{\eta}_n]\,\mathrm{e}^{2i\theta(\widehat{\eta}_n)}\widehat{D}_n$, $w_n^{(2)}=A_{-}[\widehat{\eta}_n]\,\mathrm{e}^{2i\theta(\widehat{\eta}_n)}$, $v_n^{(1)}=-\frac{iq_{-}}{\eta_n}$, $v_n^{(2)}=-\frac{iq_{-}}{\eta_n^2}$.

For the reflectionless case, that is, $\rho(z)=\hat{\rho}(z)=0$, the trace formulae and theta condition reduce to

$$s_{11}(z)=\prod_{n=1}^{N}\frac{(z-z_n)^2(z+q_0^2/z_n^*)^2}{(z-z_n^*)^2(z+q_0^2/z_n)^2}\quad\text{for } z\in D_{+}^{f},\qquad s_{22}(z)=\prod_{n=1}^{N}\frac{(z-z_n^*)^2(z+q_0^2/z_n)^2}{(z-z_n)^2(z+q_0^2/z_n^*)^2}\quad\text{for } z\in D_{-}^{f}, \tag{79}$$

and

$$\arg(q_{+}/q_{-})=8\sum_{n=1}^{N}\arg(z_n), \tag{80}$$

respectively.

The double-pole breather-breather solutions of the focusing Hirota equation with NZBCs are displayed in Fig. 8(a1)-(b3) for $N=1$. When $q_0\to 0$, we have the double-pole bright-bright solutions of the focusing Hirota equation with ZBC (see Fig. 8(c1)-(c3)).

3. The defocusing hirota equation with NZBCs

In what follows, we study the IST and soliton solutions for the defocusing ($\sigma=1$) Hirota equation with NZBCs (2).

3.1. Direct scattering with NZBCs

Similarly to the focusing case, the asymptotic problem ($|x|\to\infty$) of the defocusing Lax pair (4) with $\sigma=1$ of the defocusing Hirota equation with NZBCs is

$$\begin{aligned}&\Phi_x=X_{\pm}\Phi, && X_{\pm}=\lim_{x\to\pm\infty}X=ik\sigma_3+Q_{\pm},\\ &\Phi_t=T_{\pm}\Phi, && T_{\pm}=\lim_{x\to\pm\infty}T=[\beta(4k^2+2q_0^2)-2\alpha k]X_{\pm},\end{aligned} \tag{81}$$

which admit the fundamental matrix solutions

$$\Phi_{bg}^{d}(x,t;k)=\begin{cases}E_{\pm}^{d}(k)\,\mathrm{e}^{i\theta(x,t,k)\sigma_3}, & k\neq\pm q_0,\\ I+\left[x+(6\beta q_0^2-2\alpha k)\,t\right]X_{\pm}(k), & k=\pm q_0,\end{cases} \tag{82}$$

where

$$E_{\pm}^{d}(k)=\begin{bmatrix}1 & \frac{iq_{\pm}}{k+\lambda}\\ -\frac{iq_{\pm}^*}{k+\lambda} & 1\end{bmatrix},\quad \theta(x,t;k)=\lambda(k)\left\{x+\left[\beta\left(4k^2+2q_0^2\right)-2\alpha k\right]t\right\},\quad \lambda^2(k)=k^2-q_0^2. \tag{83}$$

To discuss the analyticity of the Jost solutions of the Lax pair (4) with $\sigma=1$, we here need to determine the regions where Im $\lambda(k)>0\,(<0)$ in $\theta(x,t;k)$ (see, e.g., Ref. [25] for the defocusing NLS equation):

- When $q_0\neq 0$, that is, the BCs are NZBCs, $\lambda(k)$ satisfying $\lambda^2(k)=k^2-q_0^2$ in the complex plane is a doubly branched function of k with two branch points being $k=\pm q_0$ and the branch cut being the segment $q_0[-1,1]$. Let $k\pm q_0=r_{\pm}\,\mathrm{e}^{i\theta_{\pm}+2im_{\pm}\pi}$ ($r_{\pm}>0$, $\theta_{\pm}\in[-\pi,\pi)$, $m_{\pm}\in\mathbb{Z}$), then two single-valued analytical branches of the complex k-plane are defined by Sheet-I: $\lambda_{\mathrm{I}}(k)=\sqrt{r_{+}r_{-}}\,\mathrm{e}^{i\frac{\theta_{+}+\theta_{-}}{2}}$ and Sheet-II: $\lambda_{\mathrm{II}}(k)=-\lambda_{\mathrm{I}}(k)$.
- In this case, we introduce a uniformization variable z defined by the conformal mapping [3]: $z=k+\lambda(k)$, whose inverse mapping is $k(z)=\frac{1}{2}\left(z+\frac{q_0^2}{z}\right)$, $\lambda(z)=\frac{1}{2}\left(z-\frac{q_0^2}{z}\right)$. The mappings between the two-sheeted Riemann k-surface and complex z-plane are given in Table 2 (see also Fig. 9).

Since the continuous spectrum of $X_{\pm}$ is the set of all values of z satisfying $\lambda(z)\in\mathbb{R}$ [28], that is, $z\in\mathbb{R}$ for which $k\in\mathbb{R}\backslash(-q_0,q_0)$. Since $T_{\pm}=[\beta(4k^2+2q_0^2)-2\alpha k]X_{\pm}$, similar to the focusing case, for all $z\in\mathbb{R}$, one can simultaneously define the Jost solutions $\Phi_{\pm}(x,t;z)$ of both parts of the defocusing Lax pair (4) satisfying the boundary conditions

$$\Phi_{\pm}(x,t;z)=E_{\pm}^{d}(z)\,\mathrm{e}^{i\theta(x,t;z)\sigma_3}+o(1),\quad x\to\pm\infty. \tag{84}$$

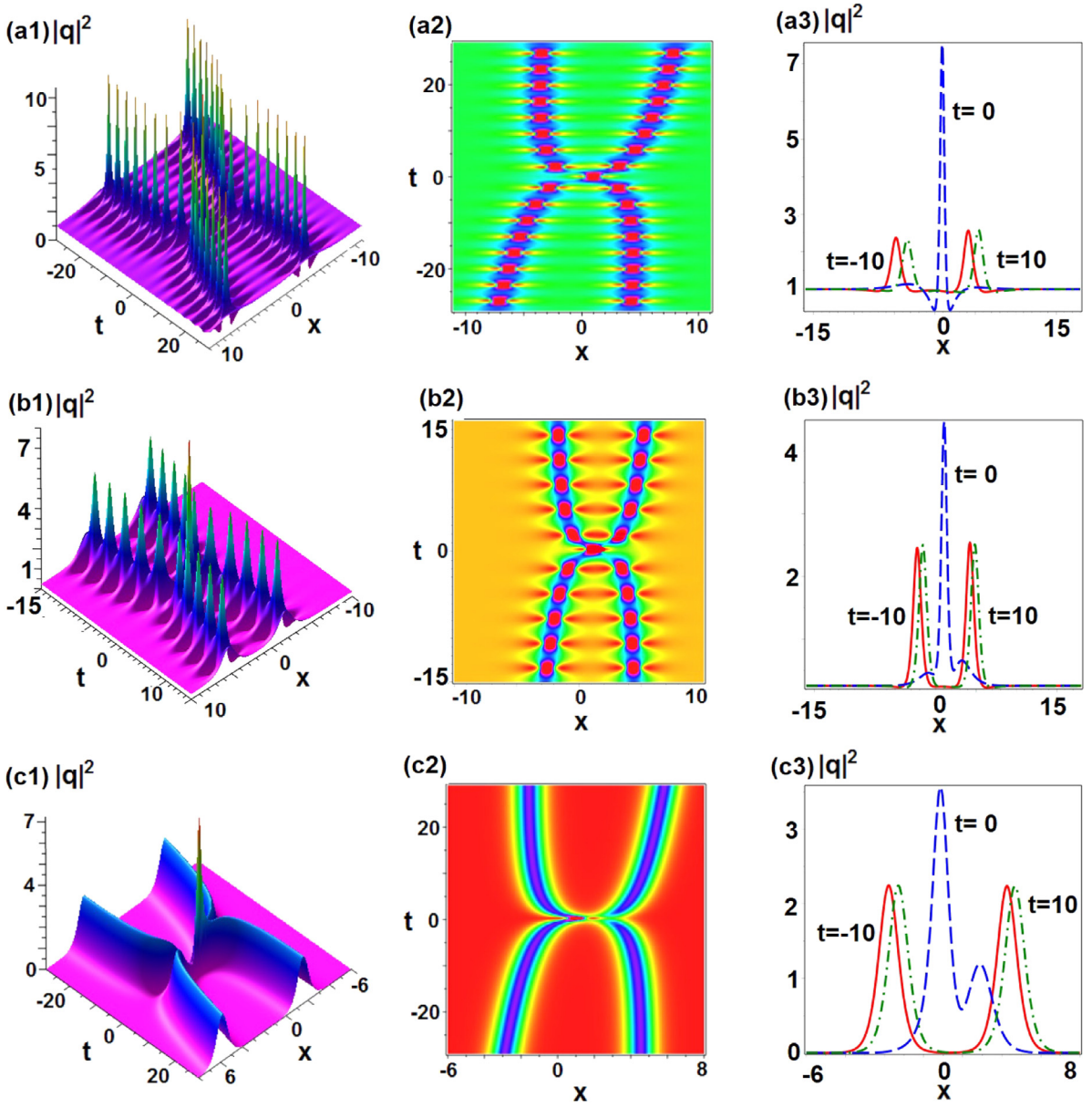


Fig. 8. Double-pole one-soliton solutions of the focusing Hirota equation with $\alpha = 1$, $\beta = 0.01$, $N = 1$, $z_1 = 3i/2$, $A_+[z_1] = B_+[z_1] = 1$. (a1, a2, a3): the breather-breather solutions with NZBCs and parameter $q_- = 1$; (b1, b2, b3): the breather-breather solutions with NZBCs and parameter $q_- = 0.5$; (c1, c2, c3): the bright-bright solitons with ZBC and parameter $q_0 \to 0$.

Table 2
Conformal mapping from two sheets of k-plane to z-plane, where $\mathbb{Q}_j^k$ ($\mathbb{Q}_j^z$), $j = 1, 2, 3, 4$ denote the first, second, third, and fourth quadrants excluding the boundaries of k-plane (z-plane), respectively.

Sheet-I of k-plane	$\to$	z-plane	Sheet-II of k-plane	$\to$	z-plane
$\mathbb{Q}_j^k$, $j = 1, 2, 3, 4$	$\to$	$\mathbb{Q}_j^z \bigcap \{\|z\| > q_0\}$, $j = 1, 2, 3, 4$	$\mathbb{Q}_j^k$, $j = 1, 2, 3, 4$	$\to$	$\mathbb{Q}_j^z \bigcap \{\|z\| < q_0\}$, $j = 1, 2, 3, 4$
$q_0[-1, 1]$	$\to$	$\{z = q_0 e^{i\theta_z},\ \theta_z \in [-\pi, 0]\}$	$q_0[-1, 1]$	$\to$	$\{z = q_0 e^{i\theta_z},\ \theta_z \in [\pi, 0]\}$
$\pm q_0[1, \infty)$	$\to$	$\pm q_0[1, \infty)$	$\pm q_0[1, \infty)$	$\to$	$\pm q_0[1, 0]$
$i[0, \pm\infty)$	$\to$	$\pm i q_0[1, \infty)$	$i[0, \pm\infty)$	$\to$	$\pm i q_0[1, 0]$

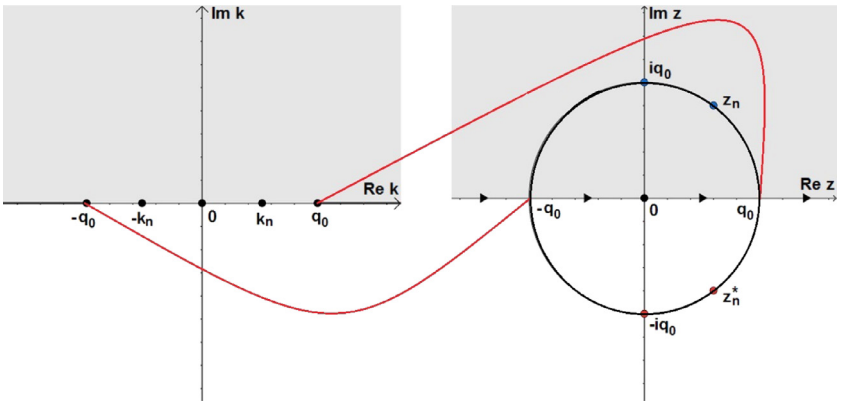


Fig. 9. Defousing Hirota equation with NZBCs. Left: the sheet-I of the Riemann surface, showing the discrete spectrum, and the regions where $\mathrm{Im}\,\lambda > 0$ (grey) and $\mathrm{Im}\,\lambda < 0$ (white). Right: the complex z-plane, showing the discrete spectrum [zeros of $s_{11}(z)$ (blue) in the grey region and those of $s_{22}(z)$ (red) in the white region], the region $\mathbb{C}^+$ where $\mathrm{Im}\,\lambda > 0$ (grey), the region $\mathbb{C}^-$ where $\mathrm{Im}\,\lambda < 0$ (white) and the orientation of the contours for the related Riemann-Hilbert problem. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Thus, the modified Jost solutions $\mu_\pm(x, t; z)$ defined by

$$\mu_\pm(x,t;z) = \Phi_\pm(x,t;z)\,\mathrm{e}^{-i\theta(x,t;z)\sigma_3} \to E^d{}_\pm(z), \quad x \to \pm\infty, \tag{85}$$

have the forms

$$\mu_\pm(x,t;z) = \begin{cases} E^d_\pm(z)\Big\{I + \int_{\pm\infty}^{x} \mathrm{e}^{i\lambda(z)(x-y)\widehat{\sigma}_3}\Big[(E^d_\pm(z))^{-1}\Delta Q_\pm(y,t)\,\mu_\pm(y,t;z)\Big]\,\mathrm{d}y\Big\}, & z \neq \pm q_0, \\ E^d_\pm(z) + \int_{\pm\infty}^{x} [I + (x-y)(Q_\pm \pm iq_0\,\sigma_3)]\Delta Q_\pm(y,t)\,\mu_\pm(y,t;z)\,\mathrm{d}y, & z = \pm q_0, \end{cases} \tag{86}$$

where $\Delta Q_\pm(x,t) = Q(x,t) - Q_\pm$ with $Q_\pm = \lim_{x\to\pm\infty} Q|_{\sigma=1}$.

Let $\mu_\pm(x,t;z) = (\mu_{\pm1}(x,t;z),\ \mu_{\pm2}(x,t;z))$ and $\Phi_\pm = (\Phi_{\pm1}(x,t;z),\ \Phi_{\pm2}(x,t;z))$, and $\Sigma_0^d = \mathbb{R}\backslash\{\pm q_0\}$. Since they contain these terms $e^{\pm i\lambda(x-y)}$ and $\mathrm{e}^{\pm i\theta(x,t;\,z)}$, thus according to the properties of these functions in distinct domains and the definition (86) of $\mu_\pm$, as well as the relation (85) between $\mu_\pm$ and $\Phi_\pm$, we easily have the following properties (see [25] for the details):

Proposition 7. *Suppose $q - q_\pm \in L^1(\mathbb{R}^\pm)$, then the Jost functions $\Phi_{\pm2}(x,t;z))$ and the modified forms $\mu_{\pm2}(x, t; z))$ given by Eqs. (85) and (86) have unique solutions in Σ_0^d. Moreover, $\mu_{+1}(x,t;z)$, $\mu_{-2}(x,t;z)$, $\Phi_{+1}(x,t;z)$ and $\Phi_{-2}(x,t;z)$ can be analytically extended to $\mathbb{C}^+$, and continuously extended to $\mathbb{C}^+ \cup \Sigma_0^d$, whereas $\mu_{-1}(x,t;z)$, $\mu_{+2}(x,t;z)$, $\Phi_{-1}(x,t;z)$, and $\Phi_{+2}(x,t;z)$ can be analytically extended to $\mathbb{C}^-$, and continuously extended to $\mathbb{C}^- \cup \Sigma_0^d$.*

Similarly to the focusing case, there also exists a constant matrix $S(z)$ between $\Phi_\pm(x,t;z)$ to generate $\Phi_+(x,t;z) = \Phi_-(x,t;z)\,S(z)$, $x, t \in \mathbb{R}$, $z \in \Sigma_0^d$, where $S(z) = \big(s_{ij}(z)\big)_{2\times2}$. Thus one can have the similar expressions for $s_{ij}(z)$ given by Eq. (16) with $\gamma_f(z)$ being replaced by $\gamma_d(z) = 1 - q_0^2/z^2 \neq 0$ for $z \neq \pm q_0$.

Notice that one can not exclude the possible presence of zeros for the scattering coefficients $s_{11}(z)$ and $s_{22}(z)$ along Σ_0^d. To solve the Riemann-Hilbert problem during the inverse process, we here require that the potential is not of spectral singularities [41], i.e., $s_{11}(z)s_{22}(z) \neq 0$ for $z \in \Sigma_0^d$. Thus the so-called reflection coefficients are defined as $\rho(z) = s_{21}(z)/s_{11}(z)$, $\hat{\rho}(z) = s_{12}(z)/s_{22}(z)$.

3.2. The inverse scattering with NZBCs and simple poles

To study the discrete spectrum and residue conditions in the inverse scattering problem, we need to analyze the symmetries of the scattering matrix $S(z)$. It follows from the previous definitions of $k(z)$ and $\lambda(z)$ that they are of the symmetries: $k(z) = k^*(z^*)$, $k(z) = k(q_0^2/z)$, $\lambda(z) = \lambda^*(z^*)$, and $\lambda(z) = -\lambda(q_0^2/z)$, thus we have the symmetries of X, T and θ as

$$\begin{aligned} &X(x,t;z) = \sigma_1 X^*(x,t;z^*)\sigma_1, \quad X(x,t;z) = X(x,t;q_0^2/z), \\ &T(x,t;z) = \sigma_1 T^*(x,t;z^*)\sigma_1, \quad T(x,t;z) = T(x,t;q_0^2/z), \\ &\theta(x,t;z) = \theta^*(x,t;z^*), \quad \theta(x,t;z) = -\theta(x,t;q_0^2/z), \end{aligned} \tag{87}$$

where $\sigma_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is the first Pauli matrix.

Combining the above-mentioned symmetries (87), Lax pair (4), and Eq. (85), we have the following symmetries for $\Phi_\pm$ and $\mu_\pm$ as

$$\begin{aligned}&\Phi_\pm(x,t;z)=\sigma_1\Phi_\pm^*(x,t;z^*)\sigma_1,\quad \Phi_\pm(x,t;z)=\frac{i}{z}\Phi_\pm(x,t;q_0^2/z)\sigma_3Q_\pm,\\&\mu_\pm(x,t;z)=\sigma_1\mu_\pm^*(x,t;z^*)\sigma_1,\quad \mu_\pm(x,t;z)=\frac{i}{z}\mu_\pm(x,t;q_0^2/z)\sigma_3Q_\pm.\end{aligned}\tag{88}$$

By means of Eq. (88) and $\Phi_+(x,t;z)=\Phi_-(x,t;z)\,S(z)$, $x,t\in\mathbb{R}$, $z\in\Sigma_0^d$, we have the symmetries

$$S(z)=\sigma_1S^*(z^*)\sigma_1,\quad S(z)=(\sigma_3Q_-)^{-1}S(q_0^2/z)\sigma_3Q_+,\tag{89}$$

which can lead to the symmetries between $\rho(z)$ and $\hat\rho(z)$ as

$$\rho(z)=\hat\rho^*(z^*),\quad \rho(z)=-\frac{q_-^*}{q_-}\hat\rho(q_0^2/z).\tag{90}$$

Similarly to [3,25] for the defocusing NLS equation, it can be shown that the discrete spectral points of the defocusing Hirota equation are also simple. Moreover, it was shown that if $(1+|x|)^4(q-q_\pm)\in L^1(\mathbb{R}^\pm)$, then there is a finite number of discrete spectral points, all of which belong to $\{z:|z|=q_0\}$ [25]. Therefore, we give the discrete spectrum as

$$Z^d=\{z_n,\,z_n^*\}_{n=1}^N,\quad z_n^*=q_0^2/z_n,\quad \mathrm{Im}\,z_n>0.\tag{91}$$

For the case of defocusing simple poles, $A_\pm[z_0]$ defined by Eq. (26) still holds, where Z^f and $D_\pm^f$ are replaced by Z^d and $\mathbb{C}^\pm$, respectively. According to the above-mentioned symmetries, we have the symmetries of $A[z_0]$ as

$$A_+[z_0]=A_-^*[z_0^*],\quad A_+[z_0]=\frac{z_0^2}{q_-^2}A_-\left[\frac{q_0^2}{z_0}\right],\quad z_0\in Z^d\cap\mathbb{C}^+\tag{92}$$

in terms of symmetries (18) and (19), which can further lead to

$$\mathrm{Im}A_+[z_n]=\left(\mathrm{Re}(q_-^2/z_n^2)-1\right)\frac{\mathrm{Re}A_+[z_n]}{\mathrm{Im}(q_-^2/z_n^2)},\quad \mathrm{Im}(q_-^2/z_n^2)\neq0,\tag{93}$$

In particular, $A_+[z_n]\in\mathbb{R}$ for $\mathrm{Re}(q_-^2/z_n^2)=1$ and $A_+[z_n]\in i\mathbb{R}$ for $\mathrm{Re}(q_-^2/z_n^2)=-1$.

Similarly to [25], we also have the asymptotic behaviors for modified Jost solutions and scattering data, which are same as ones for the focusing case given by Eq. (30). The above-mentioned Riemann-Hilbert problem for the focusing case in Proposition 2 still holds for the defocusing case.

To solve the Riemann-Hilbert problem, subtracting out the asymptotic behavior and the simple-pole contributions, that is,

$$M_{spd}(x,t;z)=I+\frac{i}{z}\sigma_3Q_-+\sum_{n=1}^N\left[\frac{\mathrm{Res}_{z=z_n}M^+(x,t;z)}{z-z_n}+\frac{\mathrm{Res}_{z=z_n^*}M^-(x,t;z)}{z-z_n^*}\right]\tag{94}$$

from both sides of the above-mentioned jump condition $M^-=M^+(I-J)$ yields

$$M^-(x,t;z)-M_{spd}(x,t;z)=M^+(x,t;z)-M_{spd}(x,t;z)-M^+(x,t;z)J(x,t;z),\tag{95}$$

in which, $M^\pm(x,t;z)-M_{spd}(x,t;z)$ are analytic in $D_\pm^f$. Moreover, their asymptotics are both $O\left(\frac{1}{z}\right)$ as $z\to\infty$ and $O(1)$ as $z\to0$, and $J(x,t;z)$ is $O\left(\frac{1}{z}\right)$ as $z\to\infty$, and $O(z)$ as $z\to0$. Therefore, the Cauchy projectors $(P^\pm[f](z)=\frac{1}{2\pi i}\int_{\Sigma^d}\frac{f(\zeta)}{\zeta-(z\pm i0)}\,d\zeta$, where the notation $z\pm i0$ represents the limit taken from the left/right of z) and Plemelj's formulae are used to solve Eq. (95) to yield

$$M(x,t;z)=M_{spd}(x,t;z)+\frac{1}{2\pi i}\int_{\Sigma^d}\frac{M^+(x,t;\zeta)J(x,t;\zeta)}{\zeta-z}\,\mathrm{d}\zeta,\quad z\in\mathbb{C}\backslash\Sigma^d,\tag{96}$$

where $\int_{\Sigma^d}$ stands for the integral along the oriented contours shown in Fig. 1(right).

It follows from Eq. (31) that only its first (second) column has a simple pole at $z=z_n$ ($z=z_n^*$). Therefore, by means of Eqs. (12) and (26), the residue parts in Eq. (96) can be found as

$$\frac{\mathrm{Res}_{z=z_n}M^+(x,t;z)}{z-z_n}+\frac{\mathrm{Res}_{z=z_n^*}M^-(x,t;z)}{z-z_n^*}=\left(\frac{A_+[z_n]\,\mathrm{e}^{-2i\theta(x,t;z_n)}}{z-z_n}\mu_{-2}(x,t;z_n),\frac{A_-[z_n^*]\,\mathrm{e}^{2i\theta(x,t;z_n^*)}}{z-z_n^*}\mu_{-1}(x,t;z_n^*)\right),\tag{97}$$

For $z=z_s$ $(s=1,2,\cdots,N)$, it follows from the second column of $M(x,t;\lambda)$ given by Eq. (96) with Eq. (97) that

$$\mu_{-2}(x,t;z_s)=\begin{bmatrix}\frac{iq_-}{z_s}\\1\end{bmatrix}+\sum_{n=1}^N\frac{A_-[z_n^*]\,\mathrm{e}^{2i\theta(x,t;z_n^*)}}{z_s-z_n^*}\mu_{-1}(x,t;z_n^*)+\frac{1}{2\pi i}\int_{\Sigma^d}\frac{(M^+J)_2(x,t;\zeta)}{\zeta-z_s}\,d\zeta,\quad s=1,2,\cdots,N.\tag{98}$$

Moreover, it follows from Eq. (18) that

$$\mu_{-2}(x,t;z)=\frac{iq_-}{z}\,\mu_{-1}\big(x,t;q_0^2/z\big), \tag{99}$$

which further leads to

$$\mu_{-2}(x,t;z_s)=\frac{iq_-}{z_s}\,\mu_{-1}\big(x,t;q_0^2/z_s\big)=\frac{iq_-}{z_s}\,\mu_{-1}(x,t;z_s^*),\quad z_s\in Z^d,\quad s=1,2,\ldots,N. \tag{100}$$

The substitution of Eq. (100) into Eq. (98) yields

$$\sum_{n=1}^{N}\left(\frac{A_-[z_n^*]\,\mathrm{e}^{2i\theta(x,t;z_s^*)}}{z_s-z_n^*}-\frac{iq_-}{z_s}\,\delta_{sn}\right)\mu_{-1}(x,t;z_n^*)+\begin{bmatrix}\frac{iq_-}{z_s}\\1\end{bmatrix}+\frac{1}{2\pi i}\int_{\Sigma^d}\frac{(M^+J)_2(x,t;\zeta)}{\zeta-z_s}\,d\zeta=0,\quad s=1,2,\cdots,N, \tag{101}$$

where δ_{sn} is the Kronecker delta function. The system (101) containing N equations with N unknowns $\mu_{-1}(x,t;z_n^*), n=1,2,\cdots,2N$ can yield the solutions $\mu_{-1}(x,t;z_s^*), s=1,2,\cdots,2N$ such that one can also get $\mu_{-2}(x,t;z_s^*), s=1,2,\cdots,N$ from Eq. (100). Therefore, substituting the obtained $\mu_{-1}(x,t;z_s^*)$ and $\mu_{-2}(x,t;z_s), s=1,2,\cdots,N$ into Eq. (97), and then substituting Eq. (97) into Eq. (96) can yield the $M(x,t;z)$ by means of the scattering data.

It follows from Eq. (35) with Eq. (36) that the asymptotic behavior of $M(x,t;z)$ as

$$M(x,t;z)=I+\frac{M^{(1)}(x,t)}{z}+O\Big(\frac{1}{z^2}\Big),\quad z\to\infty, \tag{102}$$

where

$$\begin{aligned}M^{(1)}(x,t)=&\,i\sigma_3\,Q_-+\sum_{n=1}^{N}\Big[A_+[z_n]\,\mathrm{e}^{-2i\theta(x,t;z_n)}\mu_{-2}(x,t;z_n),\,A_-[z_n^*]\,\mathrm{e}^{2i\theta(x,t;z_n^*)}\mu_{-1}(x,t;z_n^*)\Big]\\&-\frac{1}{2\pi i}\int_{\Sigma^d}M^+(x,t;\zeta)\,J(x,t;\zeta)\,\mathrm{d}\zeta.\end{aligned} \tag{103}$$

with $\mu_{-1}(x,t;z_s^*)$ and $\mu_{-2}(x,t;z_s), s=1,2,\cdots,2N$ given by Eqs. (101) and (100).

It follows from Eq. (31) that $M(x,t;z)e^{i\theta(x,t;z)\sigma_3}$ satisfies the Lax pair (4). Substituting $M(x,t;z)e^{i\theta(x,t;z)\sigma_3}$ with Eq. (40) into the x-part of the Lax pair (4) and then choosing the coefficients of z^0 yield the following property for the potential $q(x,t)$:

Proposition 8. *The potential with simple poles of the defocusing Hirota equation with NZBCs is given by*

$$q(x,t)=q_--i\sum_{n=1}^{N}A_-[z_n^*]\,\mathrm{e}^{2i\theta(x,t;z_n^*)}\mu_{-11}(x,t;z_n^*)+\frac{1}{2\pi}\int_{\Sigma^d}\big(M^+J\big)_{12}(\zeta)\,\mathrm{d}\zeta, \tag{104}$$

where $\mu_{-11}(x,t;z_n^*)$'s are determined by

$$\frac{iq_-}{z_s}+\sum_{n=1}^{N}\left(\frac{A_-[z_s^*]\,\mathrm{e}^{2i\theta(x,t;z_s^*)}}{z-z_s^*}-\frac{iq_-}{z_s}\,\delta_{sn}\right)\mu_{-11}(x,t;z_n^*)+\frac{1}{2\pi i}\int_{\Sigma^d}\frac{(M^+J)_{12}(\zeta)}{\zeta-z_s}\,\mathrm{d}\zeta=0,\ s=1,2,\ldots,N, \tag{105}$$

which are deduced from Eq. (101).

Similarly to the case of focusing simple poles, we also have the trace formulae for the case of defocusing simple poles

$$s_{11}(z)=e^{s(z)}s_0(z)\ \text{ for }\ z\in\mathbb{C}^+,\quad s_{22}(z)=e^{-s(z)}/s_0(z)\ \text{ for }\ z\in\mathbb{C}^-, \tag{106}$$

where $s(z)=\frac{i}{2\pi}\int_{\Sigma^d}\frac{\log[1+\rho(\zeta)\,\rho^*(\zeta^*)]}{\zeta-z}\,d\zeta$ and $s_0(z)=\prod_{n=1}^{N}\frac{z-z_n}{z-z_n^*}$.

According to the limit $z\to 0$ of $s_{11}(z)$ in Eq. (106) and Eq. (30), one can yield the theta condition as

$$\arg(q_+/q_-)=2\sum_{n=1}^{N}\arg(z_n)+\int_{\Sigma^d}\frac{\log[1+\rho(\zeta)\,\rho^*(\zeta^*)]}{2\pi\zeta}\,d\zeta. \tag{107}$$

In particular, for the reflectionless case, that is, $\rho(z)=\hat{\rho}(z)=0$, which leads to $J=0$, we have the following proposition:

Proposition 9. *The reflectionless potential with simple poles of the defocusing Hirota equation with NZBCs is given by*

$$q(x,t)=q_-+i\frac{\det\begin{bmatrix}H & v\\ w^T & 0\end{bmatrix}}{\det H}, \tag{108}$$

where $H=(h_{kn})_{N\times N}$, $w=(w_n)_{N\times 1}$, $v=(v_k)_{N\times 1}$, $h_{kn}=w_k/(z-z_k^*)-\frac{iq_-}{z_k}\,\delta_{k,n}$, $w_n=A_-[z_n^*]\,\mathrm{e}^{2i\theta(x,t;z_n^*)}$, and $v_k=-iq_-/z_k$.

For the reflectionless case, the trace formulae and theta condition can reduce to

$$s_{11}(z)=\prod_{n=1}^{N}\frac{z-z_n}{z-z_n^*}\quad\text{for}\quad z\in\mathbb{C}^+,\quad s_{22}(z)=\prod_{n=1}^{N}\frac{z-z_n^*}{z-z_n}\quad\text{for}\quad z\in\mathbb{C}^-, \tag{109}$$

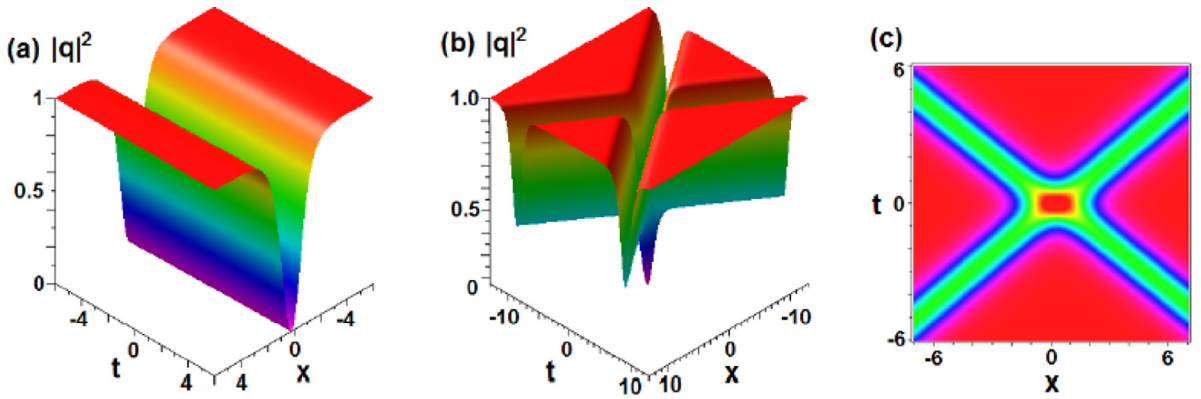


Fig. 10. Simple-pole solitons of the defousing Hirota equation with NZBCs and $\alpha = 1$, $\beta = 0.01$. (a): one-dark soliton with parameters: $N = 1$, $q_- = 1$, $z_1 = i$, $\mathrm{Im}A_+[z_1] = 1$; (b)-(c) dark-dark soliton with parameters: $N = 2$, $q_- = 1$, $z_1 = \mathrm{e}^{\frac{\pi}{4}i}$, $z_2 = \mathrm{e}^{\frac{3\pi}{4}i}$, $\mathrm{Re}A_+[z_1] = 1$, $\mathrm{Re}A_+[z_2] = -1$.

and

$$\arg(q_+/q_-) = 2\sum_{n=1}^{N}\arg(z_n), \tag{110}$$

respectively.

For $N = 1$, $\alpha = 1$, $\beta = 0.01$, $q_- = 1$, $z_1 = i$, $\mathrm{Im}A_+[z_1] = 1$, we have the simple-pole one-dark of the defocusing Hirota equation with NZBCs from Proposition 9

$$q(x,t) = -\tanh\left(x + \frac{1}{50}t - \frac{1}{2}\ln(2)\right), \tag{111}$$

which is displayed in Fig. 10(a). The phase difference is π.

For $N = 2$, $\alpha = 1$, $\beta = 0.01$, $q_- = 1$, $z_1 = \mathrm{e}^{\frac{\pi}{4}i}$, $z_2 = \mathrm{e}^{\frac{3\pi}{4}i}$, $\mathrm{Re}A_+[z_1] = 1$, $\mathrm{Re}A_+[z_2] = -1$, we have the simple-pole two-dark-dark solitons of the defocusing Hirota equation with NZBCs from Proposition 9

$$q(x,t) = \frac{2\sqrt{2}\cosh\left(x - \frac{1}{2}\ln(2)\right) + i\sinh(2t)}{2\sqrt{2}\cosh\left(x - \frac{1}{2}\ln(2)\right) + 3\cosh(2t)}, \tag{112}$$

which is displayed in Fig. 10(b) and (c). The phase difference is 2π.

For $N > 2$, we can also find the multi-dark solitons of the defocusing Hirota equation with NZBCs (2) from Eq. (108), which are omitted here.

Remark 1. These results can reduce to ones of the focusing NLS equation with NZBCs or ZBC for $\beta = 0$ [26,30] and focusing complex mKdV equation with NZBCs or ZBC for $\alpha = 0$, and ones of the defocusing NLS equation with NZBCs for $\beta = 0$ [25] and defocusing complex mKdV equation with NZBCs for $\alpha = 0$.

Remark 2. Similarly to the NLS equation, the focusing Hirota equation with NZBCs exhibits modulation instability, but the defocusing Hirota equation with NZBCs does not have this phenomenon [39,42].

4. Conclusions and discussions

In conclusion, we have presented the ISTs for both focusing and defocusing Hirota equations with NZBCs at infinity. Based on the matrix Riemann-Hilbert problems and different types of poles, we have obtained the solutions of the focusing and defocusing Hirota equations with NZBCs. Particularly, we give the explicit expressions for the reflectionless potentials in both focusing and defocusing cases. Moreover, we exhibit some simple-pole and double-pole solutions for the focusing case, and the simple-pole solutions for the defocusing case. These results can also reduce to ones of the both focusing and defocusing NLS equations and complex mKdV equations with NZBCs. Of course, the idea used in this paper can also be extended to study the ISTs and solutions for the focusing and defocusing mKdV equation [43], nonlocal mKV equation [44], derivative NLS equation [45], higher-order NLS equation [37], or other nonlocal NLS equations [14,46,47] with NZBCs.

Acknowledgements

The authors thank the referees for their valuable suggestions. This work was partially supported by the NSFC under grants Nos. 11731014 and 11571346, and CAS Interdisciplinary Innovation Team.

Appendix A. The trace formulae

Since $s_{11}(z)$ and $s_{22}(z)$ are analytic in D_+^f and D_-^f, respectively, and the discrete spectral points η_n's and $\widehat{\eta}_n$'s are the simple zeros of $s_{11}(z)$ and $s_{22}(z)$, respectively, then one can introduce [26]

$$f^+(z) = s_{11}(z)\prod_{n=1}^{2N}\frac{z-\widehat{\eta}_n}{z-\eta_n}, \qquad f^-(z) = s_{22}(z)\prod_{n=1}^{2N}\frac{z-\eta_n}{z-\widehat{\eta}_n}. \tag{113}$$

One can show that $f^{\pm}(z)$ are analytic and have no zeros in $D_{\pm}^f$. Since $\det S(z) = s_{11}(z)\,s_{22}(z) - s_{12}(z)\,s_{21}(z) = 1$, that is, $s_{11}(z)\,s_{22}(z) = 1 + s_{11}(z)\,s_{22}(z)\rho(z)\,\hat{\rho}(z)$, thus we have $f^+(z)\,f^-(z) = s_{11}(z)s_{22}(z) = \frac{1}{1-\rho(z)\,\hat{\rho}(z)}$, which leads to

$$-\log f^+(z) - \log f^-(z) = \log[1-\rho(z)\,\hat{\rho}(z)] = \log[1+\rho(z)\,\rho^*(z^*)] \tag{114}$$

in terms of $\hat{\rho}(z) = -\rho^*(z^*)$ from Eq. (20).

Since it follows from Eqs. (19) and (113) that $f^{\pm}(z)\to 1$ as $z\to\infty$, thus $\log f^{\pm}(z)\to 0$ as $z\to\infty$. Applying the Cauchy projectors $P^{\pm}$ and Plemelj's formulae to Eq. (114) to yield

$$\log f^{\pm}(z) = \mp\frac{1}{2\pi i}\int_{\Sigma^f}\frac{\log[1+\rho(\zeta)\,\rho^*(\zeta^*)]}{\zeta - z}\,d\zeta, \quad z\in D_{\pm}^f. \tag{115}$$

Therefore, the trace formulae are given in Eq. (44).

References

[1] Gardner CS, Greene JM, Kruskal MD, Miura RM. Method for solving the Korteweg-de Vries equation. Phys Rev Lett 1967;19:1095–7.
[2] Ablowitz MJ, Segur H. Solitons and the inverse scattering transform. Philadelphia: SIAM; 1981.
[3] Faddeev LD, Takhtajan LA. Hamiltonian methods in the theory of solitons. Berlin: Springer; 1987.
[4] Ablowitz AJ, Clarkson PA. Solitons, nonlinear evolution equations and inverse scattering. Cambridge: Cambridge Univeristy Press; 1991.
[5] Agrawal GP. Nonlinear fiber optics. the fifth edition. New York: Academic Press; 2012.
[6] Kivshar YS, Agrawal GP. Optical solitons: from fibers to photonic crystals. New York: Academic Press; 2013.
[7] Malomed BA, Mihalache D, Wise F, Torner L. Spatiotemporal optical solitons. J Opt B 2005;7:R53.
[8] Pitaevskii L, Stringari S. Bose-Einstein condensation and superfluidity. Oxford: Oxford University Press; 2016.
[9] Osborne AR. Nonlinear ocean waves. New York: Academic; 2009.
[10] Yan Z. Financial rogue waves. Commun Theor Phys 2010a;54:947–9.
[11] Yan Z. Vector financial rogue waves. Phys Lett A 2010b;375:4274–9.
[12] Yan Z. Nonautonomous 'rogons' in the inhomogeneous nonlinear Schrödinger equation with variable coefficients. Phys Lett A 2010c;374:672–9.
[13] Yan Z. Localized analytical solutions and parameters analysis in the nonlinear dispersive Gross-Pitaevskii mean-field GP(m, n) model with space-modulated nonlinearity and potential. Stud Appl Math 2014;132:266–84.
[14] Yan Z. Integrable PT-symmetric local and nonlocal vector nonlinear Schrödinger equations: a unified two-parameter model. Appl Math Lett 2015;47:61.
[15] Zakharov VE, Shabat AB. Exact theory of two-dimensional self-focusing and one-dimensional self-modulation of waves in nonlinear media. Sov Phys JETP 1972;34:62–9.
[16] Zakharov VE, Shabat AB. Interaction between solitons in a stable medium. Sov Phys JETP 1973;37:823–8.
[17] Kuznetsov EA. Solitons in a parametrically unstable plasma. Sov Phys Dokl 1977;22:507–8.
[18] Ma YC. The perturbed plane-wave solutions of the cubic Schrödinger equation. Stud Appl Math 1979;60:43–58.
[19] Tajiri M, Watanabe Y. Breather solutions to the focusing nonlinear Schrödinger equation. Phys Rev E 1998;57:3510–19.
[20] Zakharov VE, Gelash AA. Nonlinear stage of modulation instability. Phys Rev Lett 2013;111:054101.
[21] Akhmediev NN, Korneev VI. Modulational instability and periodic solutions of the nonlinear Schrödinger equation. Theor Math Phys 1986;69:1089–93.
[22] Prinari B, Ablowitz MJ, Biondini G. Inverse scattering transform for the vector nonlinear Schrödinger equation with nonvanishing boundary conditions. J Math Phys 2006;47:063508.
[23] Ablowitz MJ, Biondini G, Prinari B. Inverse scattering transform for the integrable discrete nonlinear Schrödinger equation with nonvanishing boundary conditions. Inverse Prob 2007;23:1711–58.
[24] Prinari B, Biondini G, Trubatch AD. Inverse scattering transform for the multi-component nonlinear Schrödinger equation with nonzero boundary conditions. Stud Appl Math 2011;126:245–302.
[25] Demontis F, Prinari B, van der Mee C, Vitale F. The inverse scattering transform for the defocusing nonlinear Schrödinger equations with nonzero boundary conditions. Stud Appl Math 2013;131:1–40.
[26] Biondini G, Kovačič G. Inverse scattering transform for the focusing nonlinear Schrödinger equation with nonzero boundary conditions. J Math Phys 2014;55:031506.
[27] Prinari B, Vitale F. Inverse scattering transform for the focusing nonlinear Schrödinger equation with one-sided nonzero boundary condition. Cont Math 2015;651:157–94.
[28] Demontis F, Prinari B, van der Mee C, Vitale F. The inverse scattering transform for the focusing nonlinear Schrödinger equation with asymmetric boundary conditions. J Math Phys 2014;55:101505.
[29] Biondini G, Fagerstrom E, Prinari B. Inverse scattering transform for the defocusing nonlinear Schrödinger equation with fully asymmetric non-zero boundary conditions. Physica D 2016;333:117–36.
[30] Pichler M, Biondini G. On the focusing non-linear Schrödinger equation with non-zero boundary conditions and double poles. IMA J Appl Math 2017;82:131–51.
[31] Kodama Y. Optical solitons in a monomode fiber. J Stat Phys 1985;39:597.
[32] Kodama Y, Hasegawa A. Nonlinear pulse propagation in a monomode dielectric guide. IEEE J Quantum Electron 1987;23:510.
[33] Yan Z, Dai C. Optical rogue waves in the generalized inhomogeneous higher-order nonlinear Schrödinger equation with modulating coefficients. J Opt 2013;15:064012.
[34] Hirota R. Exact envelope-soliton solutions of a nonlinear wave equation. J Math Phys 1973;14:805.
[35] Ankiewicz A, Soto-Crespo JM, Akhmediev N. Rogue waves and rational solutions of the Hirota equation. Phys Rev E 2010;81:046602.
[36] Tao Y, He J. Multisolitons, breathers, and rogue waves for the Hirota equation generated by the Darboux transformation. Phys Rev E 2012;85:026601.
[37] Zhang G, Yan Z, Wang L. The general coupled Hirota equations: modulational instability and higher-order vector rogue wave and multi-dark soliton structures. Proc. R. Soc. A 2019;475:20180625.
[38] Wang DS, Chen F, Wen XY. Darboux transformations of the general Hirota equation: multisoliton solutions, breather solutions, and rogue wave solutions. Adv Diff Eq 2016;2016:67.

[39] Yang Y, Yan Z, Malomed BA. Rogue waves, rational solitons, and modulational instability in an integrable fifth-order nonlinear Schrödinger equation. Chaos 2015b;25:103112.
[40] Chen S, Yan Z. The Hirota equation: Darboux transform of the Riemann-Hilbert problem and higher-order rogue waves. Appl Math Lett 2019;95:65.
[41] Zhou X. Direct and inverse scattering transforms with arbitrary spectral singularities. Commun Pure Appl Math 1989;42:895–938.
[42] Achilleos V, Diamantidis S, Frantzeskakis1 DJ, Karachalios NI, Kevrekidis PG. Conservation laws, exact traveling waves and modulation instability for an extended nonlinear Schrödinger equation. J Phys A 2015;48:355205.
[43] Zhang G., Yan Z. Inverse scattering transforms and solutions for the focusing and defocusing mKdV equations with non-zero boundary conditions. 2018a. ArXiv:1810.12150.
[44] Zhang G., Yan Z. A unified inverse scattering transform for the nonlocal modified KdV equation with non-zero boundary conditions. 2018b. ArXiv:1810.12143.
[45] Zhang G., Yan Z. Inverse scattering transforms and n-double-pole solutions for the derivative NLS equation with zero/non-zero boundary conditions. 2018c. ArXiv:1812.02387.
[46] Yan Z. Nonlocal general vector nonlinear Schrödinger equations: integrability, PT symmetribility, and solutions. Appl Math Lett 2016;62:101.
[47] Yan Z. A novel hierarchy of two-family-parameter equations: local, nonlocal, and mixed-local, nonlocal vector nonlinear Schrödinger equations. Appl Math Lett 2018;79:123.